Legal AI: From Case-Based Reasoning to Reliable, Structure-Aware Retrieval-Augmented Generation

Ali Hamza

Department of Software Engineering, Military College of Signals (MCS),

National University of Sciences and Technology (NUST), Islamabad, Pakistan

ahamza.msse25mcs@student.nust.edu.pk

October 22, 2026

Abstract

Legal work depends on finding, interpreting, and applying authoritative texts. Lawyers and judges work with statutes, regulations, prior decisions, and procedural rules. A useful AI system must therefore do more than produce fluent language: it must connect its claims to relevant and reliable legal sources. This survey traces the development of legal artificial intelligence from case-based reasoning in the early 1990s to contemporary legal language models and retrieval-augmented generation (RAG). Rather than presenting a simple chronology, we explain how each wave of methods responded to limitations in earlier approaches. We review foundational reasoning systems, legal language models, benchmarks, legal information retrieval, RAG, and studies of hallucination and citation reliability. We then examine three recent system designs that improve different parts of the pipeline: document drafting with RAG, fine-tuning of retrieval and generation, and structure-aware indexing. The review highlights persistent challenges in statutory hierarchy preservation, citation verification, multilingual access, and jurisdictional coverage, with particular attention to under-represented legal systems such as Pakistan. We distinguish findings reported by the original authors from limitations inferred from a study's scope or evaluation.

1 Introduction

Law is a domain in which the quality of an answer depends not only on whether it sounds reasonable, but also on whether it is supported by the correct legal authority. A useful legal information system must identify relevant statutes and precedents, interpret them in context, account for their relationships, and present conclusions that can be checked against authoritative sources. These requirements make legal artificial intelligence (LegalAI) a distinctive research challenge. Unlike many general question-answering tasks, legal information retrieval can depend on exact statutory language, the relationship between a provision and its subordinate clauses, the authority of a judicial decision, and the jurisdiction and date to which a legal rule applies. A response that overlooks one of these conditions may appear convincing while remaining legally incomplete or incorrect.

The effort to automate legal reasoning predates modern machine learning. Early research explored how computational systems could represent legal rules, compare cases, and construct arguments resembling those used by lawyers. Ashley's work on case-based reasoning, including HYPO, demonstrated how a system could reason about legal disputes by comparing fact patterns and identifying legally meaningful similarities and differences (Ashley, 1991). This tradition established an important principle: legal reasoning cannot always be reduced to matching a question with a sentence. It often requires identifying which facts matter, which rules apply, and why one authority supports an argument more strongly than another. However, rule-based and case-based approaches also faced practical constraints. Their performance depended on how legal knowledge was represented, how exceptions and changing authorities were handled, and how much domain-specific knowledge could be encoded and maintained.

The growth of statistical natural language processing and pretrained language models changed how researchers approached these problems. Rather than relying exclusively on manually encoded rules, machine-learning systems began to learn linguistic patterns from legal documents and perform tasks such as document classification, information extraction, and case retrieval. Domain-adapted models such as LEGAL-BERT illustrated the value of training language representations on legal text rather than relying only on general-purpose language data (Chalkidis et al., 2020). Benchmarks such as LexGLUE and LegalBench subsequently provided more systematic ways to evaluate legal language models across different tasks and forms of legal reasoning (Chalkidis et al., 2022; Guha et al., 2023). These developments helped move LegalAI from narrowly engineered expert systems toward more flexible, data-driven models.

The arrival of large language models (LLMs) expanded these capabilities further. Modern models can summarize legal documents, explain provisions in natural language, compare arguments, and draft responses

to complex questions. Yet fluency introduces a serious reliability problem. A model may generate an answer that is coherent and professionally written without accurately reflecting the law, may cite a nonexistent authority, or may combine correct legal statements in a way that produces an unsupported conclusion. Research on legal hallucinations has shown why apparent linguistic quality cannot be treated as evidence of legal correctness (Dahl et al., 2024; Magesh et al., 2025). In legal practice, this distinction is particularly important because users need to verify not only the answer, but also the authorities and reasoning on which it depends.

Retrieval-augmented generation (RAG) offers one response to this challenge. Instead of asking an LLM to answer exclusively from its learned parameters, a RAG system retrieves external documents and supplies their content as evidence for answer generation. This design can make legal answers more current, traceable, and specific to an authoritative corpus. However, retrieval does not automatically guarantee correctness. If a system retrieves an irrelevant judgment, misses a decisive statutory exception, or separates a provision from its legal hierarchy, the generator may produce a plausible answer from incomplete evidence. Likewise, the presence of a citation does not establish that the cited passage actually supports the associated claim. Reliable legal RAG therefore depends on the interaction between retrieval quality, document representation, evidence selection, generation, and verification, not on the language model alone.

A central challenge is that legal documents are structured sources of authority, not merely collections of text passages. Statutes contain parts, chapters, sections, articles, clauses, provisos, and exceptions; judgments contain facts, issues, arguments, reasoning, and decisions. Flattening these documents into independent text chunks can remove the context needed to interpret a retrieved passage correctly. A short provision may appear relevant when read in isolation but mean something different when its parent section, exception, amendment, or related provision is considered. Structure-aware indexing attempts to preserve this context by attaching information about a passage's position in the original document. Hybrid retrieval, which combines lexical matching with semantic similarity, addresses a complementary problem: exact legal expressions and section numbers can be important, while conceptually relevant evidence may use different wording from the user's question. Rank-fusion methods can combine these complementary retrieval signals, although their effectiveness must be evaluated rather than assumed.

Recent research reflects the growing breadth of this field. Broad surveys now examine LegalAI through the roles of data, modelling, retrieval, generation, and evaluation, while more focused studies investigate case retrieval, legal information extraction, and the reliability of legal language models (Feng et al., 2024; Premasiri et al., 2025; Kelsall et al., 2025; Chi et al., 2026). These reviews provide essential foundations, but the field still benefits from a connected account of how legal AI's design priorities have evolved: from representing legal reasoning, to learning legal language, to retrieving external evidence, and finally to preserving the structure and provenance needed to verify generated answers.

This survey follows that evolution. Rather than treating every model or benchmark as an isolated development, it examines the relationship between legal representation, retrieval architecture, evidence grounding, and reliability. It considers classical case-based reasoning, legal language models and benchmarks, legal RAG, structure-aware and hybrid retrieval, hallucination and citation verification, and the challenges created by jurisdictional and linguistic variation. Particular attention is given to systems that must operate over structured legal authorities and to jurisdictions that are less represented in dominant English-language datasets. Pakistani legal information retrieval provides a useful motivating context because its authoritative sources include constitutional provisions, legislation, procedural laws, and judicial decisions whose relationships must be preserved during retrieval. It is considered here as a specific instance of a wider challenge: a legal AI system cannot be considered reliable merely because its language model is powerful if its evidence representation fails to preserve the legal context in which that evidence must be interpreted.

The survey makes four contributions:

1. **Historical synthesis:** It connects early computational legal reasoning with domain-specific language models and contemporary retrieval-augmented systems, explaining the problems that motivated each transition.

2. **Architectural analysis:** It distinguishes document representation, retrieval, rank fusion, generation, and evidence verification, clarifying how these components contribute to the overall reliability of legal AI.

3. **Critical comparison:** It examines contemporary legal AI systems in terms of their design choices, evaluation methods, reported strengths, and limitations, while distinguishing results obtained under different datasets and experimental conditions.

4. **Research agenda:** It identifies open challenges in structure-preserving retrieval, jurisdiction-aware evidence selection, multilingual legal access, knowledge-graph integration, and verifiable answer generation.

These contributions are organized around the research questions in Table 1.

Together, these questions frame the survey around a single problem: how can legal AI move beyond generating plausible legal language toward producing answers whose supporting evidence is relevant, contextually correct, and independently verifiable? The next section explains how the literature was identified, assessed, and organized to investigate this question.

Table 1: Research questions guiding the survey.

Research question	Focus	Purpose
RQ1. How has legal AI evolved?	From case-based reasoning and rule-based systems to legal language models and RAG	Identify the principal methodological transitions and the limitations that motivated them.
RQ2. How do legal retrieval architectures affect evidence quality?	Lexical, dense, hybrid, rank-fusion, structure-aware, and knowledge-graph-based retrieval	Explain how systems locate relevant authorities and preserve the context needed to interpret them.
RQ3. How should reliability be assessed in legal AI?	Hallucination, factual support, citation correctness, retrieval quality, and human evaluation	Distinguish fluent generation from evidence-supported legal answers and examine the limits of common evaluation practices.
RQ4. What challenges remain for jurisdiction-aware and multilingual legal AI?	Legal hierarchy, jurisdictional variation, English and low-resource languages, and evidence provenance	Identify research opportunities for systems that must retrieve and verify legal information across different legal and linguistic contexts.

2 Literature Search and Survey Methodology

2.1 Review design and scope

This study adopts a structured narrative review to examine the evolution of legal artificial intelligence from early computational legal reasoning to contemporary retrieval-augmented generation. The review is organized around the development of methods, the problems they were designed to address, and the limitations that remain unresolved. This approach is appropriate for a field in which research spans several related but distinct areas, including legal knowledge representation, natural language processing, information retrieval, large language models, and evidence verification.

The purpose is not simply to count publications or rank models by reported performance. Instead, the review seeks to explain how changes in system design have influenced the way legal information is represented, retrieved, interpreted, and presented to users. A chronological perspective establishes the historical progression, while a thematic analysis enables comparison across studies that address similar technical challenges using different methods.

The review also distinguishes between broad surveys of LegalAI and primary research on particular architectures or evaluation methods. Broad surveys provide context for understanding the field, whereas primary studies provide the technical details and empirical evidence needed to compare specific approaches. For example, Chi et al. (2026) organize contemporary LegalAI research around data, modelling, and evaluation, including legal classification, retrieval, and generation. This survey uses that broader landscape as context but concentrates on the relationship between retrieval architecture, preservation of legal structure, and the reliability of generated answers.

The review is therefore guided by four analytical dimensions:

- **Historical development:** How legal reasoning systems evolved into data-driven language models and retrieval-augmented systems.
- **Technical architecture:** How legal documents are represented, indexed, retrieved, ranked, and supplied to a generator.
- **Reliability and evaluation:** How studies measure retrieval effectiveness, factual support, hallucination, citation correctness, and overall answer quality.
- **Research gaps:** Which challenges remain insufficiently addressed, particularly in structure-aware retrieval, jurisdiction-specific legal information, and multilingual access.

The scope covers foundational research and contemporary developments relevant to these dimensions. Earlier publications are included where they establish concepts or methods that continue to influence modern systems. Recent work receives particular attention because the architecture and evaluation of legal AI are changing rapidly.

2.2 Information sources and search strategy

A reproducible review requires more than collecting papers through general web searches. The sources consulted, search terms used, date of each search, and process for selecting publications should be documented so that readers can understand how the literature was assembled.

The search strategy should cover both computer science and legal informatics. Relevant studies may appear in information retrieval, natural language processing, artificial intelligence, law and technology, and interdisciplinary legal research venues. Accordingly, the recommended sources and their purposes are summarized in Table 2.

Methodological note: This is the proposed source inventory for the review protocol, not a claim that every source has already been searched. The final manuscript must identify which sources were actually consulted and report the dates of those searches.

Searches should combine terms from four groups: the legal domain, the underlying technology, retrieval and document structure, and reliability. Combining these groups helps identify both foundational studies and papers that use different terminology for related problems. Example queries for each theme are listed in Table 3.

The queries above are starting points rather than a substitute for database-specific search syntax. Boolean

Table 2: Recommended information sources for the literature search.

Information source	Primary purpose
Scopus	Discover peer-reviewed research across computer science, information systems, and legal technology.
Web of Science	Identify relevant journal and conference publications and trace citation relationships.
IEEE Xplore	Locate research on information retrieval, AI architectures, and computational methods relevant to legal systems.
ACM Digital Library	Identify research on NLP, information retrieval, human–computer interaction, and AI system design.
ACL Anthology	Locate primary research on legal NLP, language models, retrieval, and language-related evaluation.
Google Scholar	Supplement database searches, identify related work, and locate papers not readily indexed elsewhere.
Official publisher and proceedings pages	Verify publication metadata, abstracts, publication status, and available full-text records.
arXiv	Identify recent preprints and emerging technical approaches, while distinguishing preprints from peer-reviewed publications.

Table 3: Search concepts and example queries.

Search theme	Example search query
Legal AI foundations	`"legal AI" OR "computational legal reasoning" OR "case-based reasoning"`
Legal language models	`("legal NLP" OR "legal language model" OR LEGAL-BERT) AND (classification OR extraction OR reasoning)`
Legal retrieval	`("legal information retrieval" OR "legal case retrieval") AND (ranking OR retrieval OR reranking)`
Retrieval-augmented generation	`("legal RAG" OR "retrieval-augmented generation") AND (law OR legal OR statute OR judgment)`
Structure-aware retrieval	`("structure-aware retrieval" OR "hierarchical indexing" OR "document structure") AND (legal OR statute OR judgment)`
Hybrid retrieval and rank fusion	`("hybrid retrieval" OR "BM25" OR "dense retrieval" OR "reciprocal rank fusion") AND (legal OR jurisprudence)`
Reliability and hallucination	`("legal hallucination" OR "citation verification" OR "factual correctness") AND ("large language model" OR RAG)`
Knowledge graphs and multilingual law	`("legal knowledge graph" OR "knowledge graph RAG" OR "multilingual legal retrieval") AND (AI OR NLP OR retrieval)`
Legal benchmarks and evaluation	`("legal benchmark" OR LegalBench OR LexGLUE OR "legal RAG benchmark") AND (evaluation OR reliability)`

operators, quotation marks, field restrictions, and supported wildcard characters differ between platforms. Each query should therefore be adapted to the source being searched, while preserving its intended concepts.

The search should proceed in two stages. First, broad queries identify the principal research areas and their foundational publications. Second, focused queries investigate specific architectural and evaluation issues, including hierarchical document representation, hybrid retrieval, rank fusion, citation verification, and multilingual legal access. Backward citation searching, which examines the references of relevant papers, and forward citation searching, which identifies later work that cites them, can then reveal important studies missed by keyword searches.

The search period should be recorded explicitly. For a survey covering the development of legal AI through 2026, foundational work published before the modern LLM era should be included where relevant, while the contemporary search should extend to the final search date in 2026. Publications appearing after that date should not be represented as having been covered.

2.3 Inclusion and exclusion criteria

Search results can contain many superficially relevant publications that do not provide evidence useful for the survey. A consistent selection policy is therefore necessary to prevent the final discussion from becoming an unstructured collection of loosely related papers.

Studies were considered relevant to the survey's scope when they contributed to at least one of the following: computational legal reasoning, legal NLP, legal information retrieval, legal language models, RAG architectures, structure-aware document processing, evidence verification, or legal AI evaluation. Particular importance was given to publications that described their methods clearly enough to support technical analysis. The full selection criteria are listed in Table 4.

These criteria should be applied with care. For example, a preprint may contain a technically important method that has not yet undergone peer review. Excluding every preprint could omit relevant emerging research, while treating every preprint as equivalent to a validated journal article would overstate the strength of the evidence. The appropriate response is to re-

Table 4: Study selection criteria.

Criterion	Include	Exclude
Topical relevance	Studies addressing legal reasoning, legal NLP, retrieval, generation, verification, or evaluation.	Studies using AI in an unrelated domain without a meaningful connection to legal information processing.
Technical contribution	Research presenting a method, architecture, dataset, benchmark, evaluation, or substantive empirical finding.	Publications that mention legal AI only in passing and provide no relevant technical or analytical contribution.
Publication type	Peer-reviewed journal articles and conference papers; relevant books and foundational works; clearly identified preprints where they add substantial value.	Duplicate records, unsupported summaries, and non-scholarly material used as if it were primary research.
Evidence availability	Publications with accessible bibliographic records and sufficient abstract, full-text, or methodological information for the intended analysis.	Records for which neither the relevant claims nor the publication details can be adequately verified.
Evaluation relevance	Studies reporting useful evaluation procedures, datasets, metrics, or clearly described qualitative analyses.	Performance claims whose methods and evidence cannot be established, unless discussed explicitly as an evidence limitation.
Jurisdiction and language	Work from any jurisdiction or language when relevant to the review's research questions.	No jurisdiction or language is excluded solely because it is underrepresented in mainstream research.
Publication date	Foundational studies from earlier periods and relevant work available by the declared search cutoff.	Work published after the search cutoff, unless clearly identified as a later addition.

tain relevant preprints when justified and label their publication status accurately.

Similarly, a paper should not be included merely because it reports a high score. The evaluation setting, dataset, baseline, and metric must be sufficiently clear to determine what the reported result actually demonstrates. A retrieval benchmark, a legal reasoning benchmark, and a human assessment of citation correctness measure different capabilities and should not be treated as interchangeable.

2.4 Screening and source verification

The selection process should move from broad discovery to detailed examination. The first stage removes duplicate records and clearly irrelevant results. The second assesses titles and abstracts against the inclusion criteria. The third examines the full text of potentially eligible studies to determine whether they provide sufficient evidence for the survey's technical and analytical objectives.

For each included publication, bibliographic details should be checked against the publisher's page, the official conference proceedings, or another authoritative publication record. DOI metadata should be verified where a DOI exists. For preprints, the version consulted and any subsequently published version should be distinguished to avoid treating multiple versions of the same work as separate studies.

The full-text assessment should focus on what each study actually establishes. Its proposed architecture, dataset, retrieval strategy, evaluation design, reported results, and acknowledged limitations should be recorded separately. When a paper reports an improvement, the review should identify the baseline and evaluation conditions rather than reproduce the result without context.

This discipline is especially important when comparing legal RAG systems. Two systems may use the same underlying language model but differ in corpus construction, chunking, retrieval, prompting, and evaluation. Their reported results cannot automatically be attributed to a single component. Likewise, a reduction in hallucination on one test set does not establish that a system is universally more reliable across jurisdictions, legal domains, or languages.

A PRISMA-style flow diagram can make the selection process transparent by reporting records identified, duplicates removed, records screened, full texts assessed, and studies included, together with reasons for exclusion. PRISMA 2020 provides established guidance on reporting these stages and their corresponding counts. However, the use of a flow diagram does not, by itself, make a review systematic. The counts must come from the actual search and screening process. Because the current search totals have not been established here, they should be entered only after the search log has been completed.

2.5 Data extraction and literature classification

After screening, the included studies should be analyzed using a consistent extraction framework. This is necessary because the literature uses different terminology for systems that may share components, and similar terms may refer to different technical operations. For example, query-time rank fusion combines the rankings returned by retrieval methods, whereas a model-training procedure may use preference rankings to select or optimize training examples. Both involve ranking, but they serve different purposes.

For each study, the extraction process should capture its research objective, legal domain and jurisdiction, technical approach, evidence sources, evaluation method, main findings, and limitations. The extracted information should then be classified into thematic cat-

egories that align with the survey's research questions, as shown in Table 5.

The extracted evidence is synthesized through both chronological and thematic analysis. Chronological analysis explains why the field moved from hand-crafted legal reasoning toward learned language representations and retrieval-augmented generation. Thematic analysis then compares the strengths and limitations of methods addressing similar problems, even when those methods appeared in different periods.

The synthesis gives particular attention to the distinction between retrieval quality and answer reliability. Retrieval quality concerns whether relevant and authoritative evidence is found and ranked appropriately. Answer reliability additionally depends on whether the generator interprets that evidence correctly, supports its claims with the cited material, and avoids unsupported conclusions. A system may perform well on one dimension and poorly on another; the survey therefore treats these as related but distinct evaluation targets.

2.6 Comparative analysis and methodological limitations

The final stage compares representative systems using their documented design choices and evaluation evidence. The comparison considers the retrieval strategy, preservation of legal document structure, use of external evidence, generation method, verification mechanisms, and evaluation design. Quantitative results are reported only with their original experimental context.

This restriction is essential because different studies may use different legal jurisdictions, corpora, query sets, baselines, and definitions of success. For example, a benchmark measuring legal classification cannot be compared directly with a human evaluation of unsupported legal claims. Similarly, Recall@k, citation precision, and hallucination rate describe different properties of a system. Combining them into a single ranking without a justified common protocol would obscure rather than clarify system performance.

The survey also recognizes limitations in the available evidence. First, database coverage and indexing practices can cause relevant studies to be missed. Second, recent methods may be available only as preprints or may lack independent evaluation. Third, the terminology used for structure-aware retrieval, legal grounding, and agentic systems is not always consistent. Fourth, published results may be difficult to reproduce when datasets, prompts, source documents, or evaluation procedures are unavailable. Finally, a literature synthesis can identify patterns and plausible explanations, but it cannot establish that one architecture is universally superior without controlled, comparable experiments.

For these reasons, the survey distinguishes three kinds of statements throughout the paper: findings directly reported by individual studies, interpretations drawn by comparing multiple studies, and open questions for future research. This distinction helps prevent architectural descriptions from being mistaken for demonstrated empirical advantages.

The resulting methodology supports a critical narrative of the field rather than a simple catalogue of publications. It establishes a consistent basis for examining how legal AI has evolved, why modern RAG systems still face reliability challenges, and which design principles may help connect legal structure, evidence retrieval, and verifiable generation.

3 Foundations of Legal Artificial Intelligence (1991–2016)

3.1 From legal knowledge to computational reasoning

The earliest generation of LegalAI was shaped by a fundamental question: can a computer represent the knowledge and reasoning that legal professionals use to resolve disputes? At the time, the dominant challenge was not generating natural-sounding legal answers. It was expressing legal rules, facts, relationships, and arguments in a form that a computational system could process.

One influential approach was case-based reasoning, which draws on the legal practice of examining previous decisions when addressing a new dispute. Ashley's work on HYPO demonstrated how a system could compare a fact pattern with earlier cases, identify legally relevant similarities and differences, and construct arguments using those comparisons (Ashley, 1991). Rather than treating a past decision as a simple answer to a new question, this approach made the reasoning process itself part of the computational problem.

Consider a simplified example. Suppose a court must decide whether a party breached a contractual obligation. A case-based system might retrieve earlier disputes involving similar obligations, compare the parties' conduct, and identify facts that strengthen or weaken an argument. If the current dispute differs from a precedent in a legally significant way, the system must account for that difference rather than blindly reuse the earlier outcome. This illustrates both the strength and the difficulty of case-based reasoning: legal relevance depends on more than textual similarity.

A related approach was the rule-based expert system. Instead of deriving guidance primarily from comparisons with previous cases, a rule-based system represents knowledge as explicit rules and applies them to known facts. In simplified form, a rule might state that if specified conditions are satisfied, a particular legal consequence follows, subject to any applicable exceptions. Such systems can make their reasoning more explicit because the rules used to reach a conclusion are represented directly.

However, legal rules rarely operate in isolation. Their application may depend on definitions, exceptions, procedural requirements, competing authorities, and facts that remain uncertain. A system that represents a general rule but omits its exceptions can produce an incorrect conclusion even when its rule-matching procedure works exactly as designed. Similarly, a manually encoded system requires ongoing maintenance as legis-

Table 5: Literature extraction and classification framework.

Analytical category	Information to extract	Role in the survey
Historical foundations	Reasoning paradigm, representation of legal knowledge, and intended task	Explain the origins of computational legal reasoning and the problems inherited by later systems.
Legal language models	Model type, training strategy, legal domain, and downstream tasks	Explain how learned representations changed legal text processing.
Retrieval architecture	Lexical or dense retrieval, hybrid search, reranking, rank fusion, and filtering	Compare methods used to identify relevant legal evidence.
Document representation	Chunking strategy, section hierarchy, metadata, and structural relationships	Assess whether retrieved passages retain the context needed for legal interpretation.
Knowledge representation	Ontologies, knowledge graphs, entities, relationships, and provenance	Examine methods for representing explicit connections among legal authorities and concepts.
Generation and orchestration	Generator, prompting, evidence use, agents, and workflow design	Explain how retrieved information is converted into legal answers.
Reliability and verification	Hallucination, factual support, citation correctness, and verification procedures	Determine how systems assess whether their answers are supported by evidence.
Evaluation	Dataset, jurisdiction, benchmark, metrics, baselines, reviewers, and statistical methods	Establish what reported results demonstrate and where comparisons are valid.
Practical constraints	Computational requirements, latency, cost, data access, and reproducibility	Identify barriers to deploying and independently evaluating legal AI systems.
Research gaps	Limitations acknowledged by authors and unresolved problems identified through comparison	Develop the future research agenda.

lation changes and courts interpret existing provisions in new ways.

These early approaches therefore established a lasting distinction. Legal reasoning requires both a representation of legal knowledge and a method for applying that knowledge to a particular dispute. A system may possess explicit rules but fail to identify the relevant facts; it may retrieve a similar case but fail to explain why the differences matter. This distinction remains important in modern legal AI, where retrieval and generation are often treated as separate components of a larger reasoning process.

3.2 Rule-based systems and legal knowledge representation

Rule-based expert systems sought to make legal knowledge operational. Legal experts or researchers encoded domain knowledge into rules, facts, and decision procedures, allowing the system to apply that knowledge to particular inputs. This approach was attractive where a legal task could be described through a relatively stable set of conditions and consequences.

The main advantage was control. Unlike a statistical model whose internal representations may be difficult to interpret, an explicitly encoded rule can often be inspected and traced. A developer can identify which condition was satisfied and which rule contributed to the outcome. This makes rule-based approaches conceptually useful for tasks requiring transparent, predefined logic.

Their limitations become clearer when legal knowledge grows in size and complexity. Rules may interact, exceptions may override general conditions, and different authorities may apply in different circumstances. Manually encoding these relationships can become expensive, while incomplete or outdated rules can lead to systematic errors. The challenge is not merely writing more rules; it is maintaining a consistent representation of the legal system and the relationships among its components.

This led to broader work on legal knowledge representation, including ontologies and formal models of legal concepts. A legal ontology defines concepts within a domain and specifies relationships among them. For example, it may distinguish a statute from a judicial decision, represent a legal provision as belonging to a particular enactment, or connect a legal obligation with the circumstances under which it applies. Such representations can support structured querying, information integration, and reasoning over explicit relationships.

Ontologies and knowledge graphs are related but not identical. An ontology defines a conceptual vocabulary and the types of relationships that may exist; a knowledge graph uses entities and relationships to represent particular information. A graph might connect a statutory provision to an amendment, a judgment that interprets it, and a legal issue addressed by that judgment. The value of these connections depends on their accuracy, completeness, and provenance.

Nevertheless, structured knowledge representation also has costs. Legal concepts can be contested, the meaning of a provision may change through judicial interpretation, and relationships may require human expertise to establish reliably. Formal representations are therefore useful but do not remove the need to interpret legal texts and verify the authority of the underlying information.

3.3 Early legal information retrieval

A different but complementary line of research focused on finding legal documents relevant to a user's information need. Legal information retrieval is not identical

to legal reasoning. Retrieval aims to identify and rank potentially relevant documents or passages, while reasoning determines how the retrieved material should be interpreted and applied.

Traditional information retrieval methods relied heavily on lexical evidence: the words appearing in a query and the documents being searched. This is valuable in law because exact wording, statutory section numbers, party names, and established legal phrases can be decisive. A query that mentions a particular provision may require that exact authority rather than a passage that merely discusses a similar concept.

However, lexical matching can miss relevant material when the query and the authoritative document use different expressions. A user may describe a legal problem in everyday language, while the relevant judgment expresses the same issue using specialized terminology. Conversely, two documents may share many words while addressing different legal issues. This creates a recurring challenge: the words used to describe a legal problem are not always the words used in the authority that resolves it.

Case retrieval adds another complication. A relevant precedent is not necessarily one that discusses the same topic. Its value may depend on the similarity of the legally material facts, the legal issue, the reasoning, and the outcome. Research on legal case retrieval has consequently developed specialized datasets and models for identifying precedents relevant to a given case, rather than relying only on general-purpose document matching (Feng et al., 2024).

By the end of this period, the main approaches had established complementary foundations. Rule-based systems offered explicit knowledge and controlled inference; case-based reasoning emphasized comparison with prior disputes; ontologies represented legal concepts and their relationships; and information retrieval provided mechanisms for finding relevant legal material. Each addressed part of the problem, but none alone solved the full challenge of locating authoritative evidence, preserving its context, and producing a reliable answer.

3.4 Lessons from the foundational period

The importance of this period lies not in a single winning architecture, but in the problems it exposed. Legal knowledge must be represented carefully, relevant authorities must be found, and reasoning must account for the relationship between a general rule and the facts of a specific case. These requirements remain relevant even as the underlying computational methods change. Table 6 summarizes the main foundational approaches.

The approaches in Table 6 should not be interpreted as mutually exclusive or as a strict sequence in which one disappeared when another emerged. They represent different ways of addressing different parts of the legal information problem. Modern systems continue to use lexical retrieval, explicit legal rules, structured metadata, and precedent comparison, even when their generators rely on neural language models.

The next phase changed the way these systems acquired and processed linguistic knowledge. Instead of depending primarily on manually encoded rules or separately engineered features, researchers increasingly trained models to learn representations from large text collections. This shift made it possible to build more flexible legal text-processing systems, but it also introduced new questions about domain adaptation, evaluation, and the reliability of model predictions.

4 Legal Language Models and Benchmarks (2017–2023)

4.1 From manually designed rules to learned legal representations

The transition from knowledge-based systems to neural language models changed how computers processed legal text. Earlier systems depended heavily on explicit rules, predefined categories, or manually designed features. Neural models instead learned statistical representations from examples, allowing them to identify linguistic patterns that were difficult to encode manually.

This transition did not make legal knowledge representation obsolete. Rather, it changed where much of the representation was learned. A neural model could learn that particular expressions frequently occur in contracts, judgments, or legislation without a developer specifying every linguistic pattern in advance. Such representations could support tasks including document classification, legal information extraction, question answering, and case retrieval.

The difficulty was that legal language differs substantially from everyday language. Statutes use carefully defined terms; contracts contain specialized expressions; judgments distinguish between the facts, arguments, reasoning, and decisions of a case. A model trained predominantly on general text may understand the grammar of a legal sentence while failing to represent the significance of its terminology.

Pretrained transformer models offered a way to address this problem. Rather than training a language model from scratch for every legal task, researchers could adapt a pretrained model to legal documents and then fine-tune it for a specific application. This approach reduced the need to build a separate linguistic system for every task, although the resulting performance still depended on the training data, model architecture, and evaluation setting.

4.2 Domain-specific models: LEGAL-BERT

A significant development was the introduction of LEGAL-BERT by Chalkidis et al. (2020). The study investigated different strategies for adapting BERT to legal NLP tasks, including using the original pretrained model, continuing pretraining on domain-specific text, and pretraining from scratch on legal corpora. The authors released a family of legal language models intended to support research in computational law and legal technology.

Table 6: Foundational approaches to LegalAI (1991–2016).

Approach	Core idea	Main strength	Principal limitation
Case-based reasoning	Compare a current dispute with previous cases and construct arguments from relevant similarities and differences.	Supports precedent-oriented reasoning and case comparison.	Requires identifying legally meaningful similarities and differences; textual similarity alone is insufficient.
Rule-based expert systems	Encode legal knowledge as explicit rules and apply them to known facts.	Transparent, inspectable rule application within a defined domain.	Difficult to maintain at scale when rules interact, exceptions arise, and legal authorities change.
Legal ontologies	Define legal concepts and their relationships in a structured vocabulary.	Supports consistent representation and structured querying of legal knowledge.	Requires careful modelling and maintenance; formal relationships do not automatically resolve legal ambiguity.
Early legal information retrieval	Search and rank legal documents using query terms and document evidence.	Helps users navigate large collections of statutes and judicial decisions.	Lexical matching can miss conceptually relevant documents or retrieve passages without sufficient legal context.
Integrated knowledge-based systems	Combine structured knowledge, retrieval, and reasoning components.	Brings different capabilities together within a single application.	System quality depends on the completeness, consistency, and coordination of its components.

The underlying principle was straightforward: if a model must process legal documents, its training should expose it to the language and terminology found in those documents. Domain adaptation can help a model learn distinctions that are less common in general-purpose text. It also provides a more systematic starting point for multiple downstream tasks than developing a new model for each one.

However, LEGAL-BERT should not be interpreted as a complete legal reasoning system. It provides learned language representations that can support particular NLP tasks; it does not automatically establish which legal authority governs a dispute, whether a provision remains applicable, or whether a generated legal conclusion is supported by the cited text. These questions require additional retrieval, legal context, and verification mechanisms.

This distinction is important for understanding the subsequent development of legal RAG. A domain-specific language model improves how a system processes legal language, but it does not necessarily provide access to current authoritative documents at inference time. Model knowledge and retrieved legal evidence serve different purposes.

4.3 LexGLUE: evaluating legal language understanding

As legal language models became more common, another problem became apparent: how could researchers determine whether a model was genuinely effective across legal tasks rather than simply successful on a single dataset?

Chalkidis et al. (2022) addressed this problem with LexGLUE, the Legal General Language Understanding Evaluation benchmark. LexGLUE brings together multiple datasets for evaluating legal language understanding in a standardized setting. Its purpose is to compare general-purpose and legal-oriented language models across different tasks rather than relying on results from a single application. The authors reported that legal-oriented models consistently improved performance over general-purpose models across multiple tasks in their evaluation.

The importance of LexGLUE lies in its emphasis on comparative evaluation. A model that performs well on one legal classification task may not perform equally well on another task requiring different linguistic or legal distinctions. A shared benchmark makes these differences easier to investigate and provides a common basis for future experiments.

Nevertheless, benchmark performance has a defined scope. LexGLUE primarily evaluates legal language understanding on its constituent datasets. Strong results do not automatically demonstrate that a system can retrieve the correct authority for a new legal question, preserve the hierarchy of a statute, generate a legally supported explanation, or verify citations. These are related capabilities, but they are not interchangeable.

4.4 LegalBench: evaluating legal reasoning

LegalBench extended the evaluation discussion toward the kinds of reasoning tasks that legal professionals consider important. Guha et al. (2023) introduced LegalBench, a collaboratively constructed benchmark containing 162 tasks covering six types of legal reasoning. The tasks were developed through collaboration between legal and computational researchers, with legal professionals contributing to their design. The study evaluated 20 open-source and commercial LLMs and examined how the tasks correspond to established ways of describing legal reasoning.

The benchmark addressed a central limitation of evaluating legal AI through generic language benchmarks alone: a model may perform well on ordinary language understanding while struggling with tasks that require legally meaningful distinctions. LegalBench provides a more direct way to examine whether models can carry out a range of tasks associated with legal analysis.

At the same time, the existence of a legal reasoning benchmark does not establish that a model is suitable for unsupervised legal practice. Performance depends on the task definitions, examples, evaluation metrics, and tested models. A benchmark can reveal strengths and weaknesses, but it cannot cover every jurisdiction, factual scenario, procedural context, or change in the law.

LegalBench and LexGLUE therefore serve complementary purposes. LexGLUE provides a standardized evaluation framework for legal language understanding, while LegalBench focuses on a broader collection of tasks designed to examine legal reasoning. Neither benchmark, by itself, resolves the problem of grounding a generated answer in the correct, current, and authoritative legal source.

4.5 What changed, and what remained unresolved?

The developments between 2017 and 2023 established two important capabilities. First, domain-adapted pretrained models made legal language processing more flexible and reusable. Second, shared benchmarks provided more systematic ways to compare models and investigate their limitations.

However, three challenges remained. First, learned representations do not guarantee access to authoritative evidence. A model may have learned patterns from legal text but cannot be assumed to know whether a statute has subsequently been amended or which authority applies to a particular dispute. Second, benchmark success is not equivalent to legal reliability. Classification accuracy or performance on a reasoning task does not establish that every claim in a generated response is factually supported or that its citations are correct. Third, legal context remains important. A model may process a paragraph accurately while overlooking the relationship between that paragraph and its parent provision, an exception, an amendment, or a related judicial interpretation.

These limitations help explain why the next phase of LegalAI increasingly incorporated retrieval into answer generation. Rather than depending exclusively on information encoded in model parameters, retrieval-augmented systems could search an external legal corpus and provide relevant documents to the generator. The aim was to combine the language capabilities of neural models with access to explicit evidence. Table 7 summarizes the main developments of this period.

The central lesson of this period is that better legal language understanding and reliable legal assistance are related, but different, goals. Domain-specific models improved how systems processed legal text, while benchmarks made their capabilities easier to measure. Yet neither development alone guaranteed that a system could find the right authority, retain its legal context, and justify its answer with verifiable evidence.

That unresolved problem motivates the next stage of the survey: the emergence of legal retrieval-augmented generation, where retrieval becomes an explicit part of the answer-generation process.

5 The Emergence of Retrieval-Augmented Generation for Law (2020–2026)

5.1 From parametric knowledge to external legal evidence

The development of large language models created a new possibility for legal AI: instead of requiring a developer to encode every rule or training a separate model for every task, a general-purpose model could interpret a question and generate a complete response in natural language. However, this flexibility introduced a fundamental difficulty. The knowledge encoded in a model's parameters is not the same as access to an authoritative, current, and verifiable legal source.

This distinction is particularly important in law. Statutes may be amended, judicial decisions may interpret or distinguish earlier authorities, and the applicability of a legal rule may depend on jurisdiction and procedural context. A language model can generate a plausible explanation without establishing which version of a provision applies or whether a cited judgment supports its conclusion. The problem is therefore not simply that a model may lack information; it is that its generated response may not provide sufficient evidence to verify its claims.

Retrieval-augmented generation (RAG) addresses this limitation by connecting a language model to an external collection of documents. Rather than relying exclusively on its learned parameters, the system retrieves relevant material and supplies that material to the generator as contextual evidence. The generator then uses the retrieved information to formulate an answer. A basic legal RAG pipeline has three stages:

1. **Indexing:** Authoritative legal documents are collected, processed, divided into retrievable units, and stored in a searchable representation.
2. **Retrieval:** A user's question is used to identify and rank potentially relevant statutes, provisions, judgments, or passages.
3. **Evidence-conditioned generation:** The retrieved material is provided to the language model, which generates a response using the supplied context.

The distinction between parametric knowledge and external evidence is central. Model parameters contain learned patterns and information acquired during training; the retrieval component provides explicit documents that can be inspected at answer time. This separation can make a legal system easier to update and its answers easier to verify. However, it does not guarantee correctness. A generator may misunderstand retrieved evidence, combine incompatible authorities, or introduce claims that the retrieved documents do not support.

RAG should therefore be understood as a mechanism for improving access to evidence, not as an automatic

Table 7: Major developments in legal language models and evaluation (2017–2023).

Development	Main contribution	What it enabled	What it did not establish
Domain-adapted neural models	Learned representations from legal language and task-specific examples.	More flexible processing of legal text than systems dependent entirely on manually designed features.	Correct legal interpretation or access to current authoritative sources.
LEGAL-BERT (Chalkidis et al., 2020)	Investigated BERT adaptation strategies and released a family of legal language models.	Reusable pretrained representations for legal NLP research.	End-to-end legal reasoning, citation verification, or guaranteed factual correctness.
LexGLUE (Chalkidis et al., 2022)	Standardized evaluation across multiple legal language understanding datasets.	More consistent comparison of general-purpose and legal-oriented models across tasks.	Reliable retrieval and generation of evidence-grounded legal answers.
LegalBench (Guha et al., 2023)	Introduced 162 collaborative legal reasoning tasks spanning six types of reasoning.	More direct investigation of LLM capabilities on tasks designed to reflect legal reasoning.	Universal competence across jurisdictions or dependable performance in real legal practice.

guarantee of legal reliability.

5.2 The legal RAG pipeline and its principal design choices

The basic RAG architecture is relatively simple, but its effectiveness depends on several interdependent decisions. The system must determine which documents to include, how to represent their structure, how to retrieve relevant evidence, how to select the strongest passages, and how to constrain the generator's response.

Legal documents make these decisions especially consequential. A statute is not merely a sequence of independent paragraphs, and a judgment is not simply a bag of semantically related sentences. The meaning of a passage can depend on its parent section, the facts of the case, the authority of the court, and the relationship between the cited material and the legal issue.

For example, a query about a constitutional protection may retrieve a paragraph that describes the general rule while missing a proviso or an associated procedural limitation. The retrieved paragraph may be relevant but insufficient. If the generator treats it as complete evidence, it can produce an answer that sounds reasonable yet omits a legally decisive qualification.

The quality of a legal RAG system is therefore determined by the complete pipeline, not merely by the language model or embedding model. Table 8 lists its principal components.

This architecture also clarifies why retrieval quality and generation quality must be evaluated separately. A retrieval system may identify the correct provision, but the generator may misinterpret it. Conversely, a capable generator may be unable to produce a reliable answer because the retrieval component omitted the decisive evidence.

5.3 From basic RAG to more adaptive systems

Legal RAG systems can be understood as a progression in how they manage retrieval and generation. Basic systems execute a relatively fixed retrieval workflow. More advanced systems add query transformation, reranking, filtering, or iterative retrieval. Modular systems separate these functions into replaceable components, while agentic systems use language-model-driven planning or tool selection to coordinate multiple steps.

These categories are useful analytical distinctions, although individual systems may combine several of them rather than fitting neatly into a single category. Table 9 summarizes the progression.

These categories should not be interpreted as a universal maturity ranking. A carefully designed basic RAG system may outperform an unnecessarily complex agentic workflow on a narrow legal question-answering task. The relevant question is whether a more elaborate architecture produces measurable improvements in evidence quality, answer reliability, or operational efficiency.

The distinction also matters for evaluation. An iterative system may retrieve additional evidence after an initial answer, but its performance should be measured against a comparable baseline. Without controlled evaluation, it is difficult to determine whether improvements arise from query refinement, better retrieval, a stronger generator, or simply additional opportunities to produce an answer.

5.4 Representative legal RAG systems

Recent studies illustrate how retrieval-augmented generation is being adapted to different legal tasks. They also show why legal RAG should not be treated as a single architecture with one universal evaluation method.

Buffa et al. (2025), in *Enhancing Legal Document Building with Retrieval-Augmented Generation*, present JusBuild, a system designed to assist legal professionals in drafting structured legal documents. The architecture segments existing documents into functional sections, stores their vector representations, retrieves relevant precedent sections, and provides additional AI-generated suggestions. An important design choice is its human-in-the-loop approach: the system assists with drafting while leaving the final formulation under the professional user's control. The study evaluates the architecture using datasets that differ in language, document template, and judicial matter. Its focus is document construction and drafting assistance rather

Table 8: Principal components of a legal RAG architecture.

Component	**Function**	**Main legal challenge**
Legal corpus	Stores legislation, regulations, judgments, and other authoritative materials.	Establishing authority, jurisdiction, currency, and source provenance.
Document preprocessing	Extracts text and metadata from PDFs, HTML, and other formats.	Handling OCR errors, complex layouts, numbering, tables, and document boundaries.
Chunking and indexing	Divides documents into retrievable units and stores their representations.	Preserving relationships between provisions, clauses, exceptions, and parent sections.
Query processing	Interprets the user's information need and prepares it for retrieval.	Mapping everyday language to specialized legal terminology and exact statutory references.
Retrieval	Identifies and ranks candidate documents or passages.	Balancing exact legal expressions with conceptual relevance.
Reranking and evidence selection	Reorders candidates and selects material for the generator.	Distinguishing merely relevant passages from legally sufficient evidence.
Generation	Produces an answer using the query and retrieved context.	Avoiding unsupported claims and interpreting evidence within its proper context.
Verification	Checks claims and citations against the retrieved sources.	Determining whether each citation actually supports the associated claim.

Table 9: Evolution of legal RAG architectures.

Architecture	**Operating principle**	**Potential benefit**	**Main limitation**
Basic or naive RAG	Retrieve passages for a query and provide them to a generator.	Simple, interpretable baseline with explicit external context.	Sensitive to chunking, retrieval failures, and irrelevant context.
Advanced RAG	Improve preprocessing, query handling, retrieval, reranking, or context selection.	Can improve evidence relevance and reduce unnecessary context.	More components introduce additional parameters, complexity, and evaluation requirements.
Modular RAG	Organize retrieval, transformation, ranking, and generation as separable components.	Supports component-level testing and replacement.	Improvements depend on the quality of component integration.
Agentic RAG	Use an agent or multiple agents to plan, select tools, refine queries, or retrieve evidence iteratively.	Can support complex questions requiring multiple retrieval steps.	May increase latency, cost, and error propagation; additional agent steps do not guarantee better answers.

than a general legal question-answering benchmark.

Ren et al. (2026) address a different problem in *Automated Data Synthesis and Retrieval-Augmented Generation for Legal Large Language Models*. Their framework combines synthetic legal data generation with quality evaluation and optimization, including an LLM-based judge and an enhanced Elo-ranking mechanism. They evaluate the approach using LawBench, a benchmark for Chinese legal language models that covers memory, comprehension, and application tasks. Their contribution is therefore not limited to query-time retrieval: it also addresses the preparation and quality of training data for legal models.

These two studies illustrate an important distinction. JusBuild uses retrieval to provide relevant precedent sections during document drafting. Ren et al. combine synthetic data preparation, model adaptation, and RAG-oriented evaluation to improve legal LLM performance. Their objectives, architectures, and evaluation settings differ, so their reported results should not be interpreted as a direct head-to-head comparison.

A further development is the explicit evaluation of the retrieval component itself. Pipitone and Houir Alami's LegalBench-RAG (Pipitone and Houir Alami, 2024) was introduced to assess retrieval for legal RAG, emphasizing the retrieval of precise text segments rather than merely identifying relevant documents. This addresses a critical evaluation gap: a system may retrieve the correct document but fail to identify the specific passage needed to support an answer.

Together, these studies demonstrate that legal RAG research addresses several related but distinct questions: how to retrieve relevant evidence, how to support document drafting, how to improve legal models through data synthesis, and how to evaluate retrieval independently from generation.

5.5 Agentic legal AI: from retrieving evidence to coordinating legal tasks

The next development is the integration of agentic workflows into legal RAG. A conventional RAG pipeline usually follows a predefined sequence: retrieve evidence, construct a prompt, and generate an answer. An agentic system can instead use a language model to plan subtasks, choose tools, reformulate queries, inspect intermediate results, and decide whether additional retrieval is necessary.

This flexibility is relevant to legal questions that cannot be answered adequately by a single retrieval operation. Consider a question involving a statutory provision and a judicial interpretation. A system might need to identify the relevant statute, retrieve judgments interpreting it, determine whether those judgments concern comparable facts, and then synthesize the evidence. An agentic workflow can coordinate these operations as separate steps.

However, additional reasoning steps also create additional opportunities for error. An agent may begin with an incorrect assumption, select an unsuitable tool, retrieve a misleading authority, or propagate an unsupported conclusion into subsequent stages. Multiple agents do not automatically provide independent verification, especially when they share the same underlying model or evidence.

For legal applications, the important design question is therefore not how many agents a system contains, but whether its workflow improves evidence quality and makes the resulting reasoning more verifiable. Each consequential claim should remain traceable to relevant source material, and the system should be able to identify when its evidence is incomplete or contradictory. Table 10 contrasts conventional and agentic legal RAG.

The broader implication is that agentic orchestration should complement, rather than replace, reliable retrieval and evidence verification. A complex workflow built on incomplete or poorly structured legal evidence remains unreliable.

5.6 Synthesis: what legal RAG solves, and what it leaves open

Retrieval-augmented generation represents an important shift in legal AI because it provides a mechanism for connecting language generation to explicit legal sources. It can make answers more specific to a selected corpus and provide material against which claims can be checked. Its modular structure also permits retrieval, generation, and verification to be improved independently.

Yet RAG does not eliminate the underlying difficulties of legal information processing. Its output remains dependent on the quality of the corpus, the preservation of document structure, the relevance of retrieved passages, and the generator's use of that evidence. Nor does a citation automatically establish correctness: the cited passage must support the claim, and the claim must be interpreted in the appropriate legal context.

This leads to the central architectural problem addressed in the next section: how can a retrieval system identify the right legal evidence while preserving the structural and semantic context required to interpret it correctly?

6 Retrieval Architectures for Legal Intelligence

6.1 Why retrieval is a distinct legal AI problem

In a legal RAG system, the generator cannot reliably compensate for evidence that the retrieval component fails to identify. If the relevant statutory provision, exception, or judicial interpretation is absent from the retrieved context, the model may generate an incomplete answer even when its language-generation capabilities are strong.

Legal retrieval therefore requires more than finding documents that discuss the same topic. The system must identify evidence that is relevant to the user's legal question, sufficiently authoritative for the intended purpose, and detailed enough to support the answer. Depending on the query, this may require an exact

Table 10: Conventional and agentic legal RAG.

Dimension	Conventional RAG	Agentic RAG
Workflow	Predetermined retrieval and generation sequence.	Workflow may adapt to intermediate results.
Query handling	Usually retrieves from the original or a transformed query.	Can plan and execute multiple related retrieval operations.
Evidence gathering	Generally performed within a fixed retrieval step.	Can iteratively gather and compare additional evidence.
Error handling	Depends on predefined fallback and verification mechanisms.	Can attempt recovery, but may also propagate errors across steps.
Evaluation priority	Retrieval effectiveness, grounded generation, latency, and cost.	The same metrics, plus planning quality, tool selection, and error propagation.

statutory citation, a particular paragraph in a judgment, a related provision, or several authorities that must be considered together.

Research on legal case retrieval has highlighted the specialized nature of this task and the need to evaluate retrieval methods using legal-specific datasets and criteria (Feng et al., 2024). Four dimensions are especially important:

- **Lexical relevance:** Does the evidence contain the exact legal terms, names, identifiers, or expressions needed?
- **Semantic relevance:** Does the evidence address the underlying legal issue, even if it uses different wording?
- **Structural relevance:** Does the retrieved passage retain the context of the provision, document section, or judgment from which it originates?
- **Evidential sufficiency:** Does the retrieved material adequately support the intended answer, including relevant exceptions or qualifications?

These dimensions are related but not interchangeable. A passage may be semantically relevant without containing the exact provision required. An exact match may retrieve a provision without its exception. A correctly retrieved paragraph may still be insufficient to establish the legal conclusion.

6.2 Lexical retrieval: precision through exact wording

Lexical retrieval ranks documents or passages according to their textual relationship with the query. One widely used method is BM25, a probabilistic ranking function that considers term frequency, inverse document frequency, and document length.

In simplified terms, BM25 gives greater importance to query terms that are informative across the collection and evaluates how frequently those terms occur in a candidate document, with adjustments that prevent very long documents from receiving an unfair advantage merely because they contain more words.

This is useful in legal retrieval because legal questions frequently include highly specific expressions. A search for a section number, statutory phrase, party name, case citation, or defined legal term may require exact matching that a purely semantic search does not consistently prioritize.

For example, a query that refers to a particular article of a constitution may benefit from matching the article identifier directly. A dense embedding system might rank a conceptually similar passage above the exact provision if the two passages are semantically close.

Lexical retrieval also has limitations. Users may describe a legal concept in everyday language while the relevant authority uses formal terminology. A query about a right to a fair hearing, for instance, may use different wording from the authoritative provision or the language of a judgment. A lexical system can miss relevant evidence when the query and the source use different expressions.

BM25 should therefore be understood as a strong retrieval baseline, particularly for exact legal terminology, rather than a complete solution to legal information retrieval.

6.3 Dense retrieval: matching legal meaning

Dense retrieval represents queries and documents as numerical vectors, usually produced by an embedding model. The system retrieves passages whose vectors are close to the query vector under a chosen similarity or distance measure.

The advantage is that retrieval can respond to conceptual similarity even when the wording differs. A user does not necessarily need to know the exact phrase used in a statute or judgment to retrieve potentially relevant material. This is particularly useful for natural-language questions, paraphrases, and legal concepts expressed differently across documents.

However, semantic similarity is not equivalent to legal relevance. Two provisions may discuss the same broad topic but apply under different conditions. A judgment may mention a legal doctrine while rejecting its applicability to the facts. A dense retriever may rank such material highly because the language is similar, even though the material does not support the answer.

Dense retrieval also depends on the quality of the embedding model and its training data. A model may represent general semantic relationships well while failing to capture distinctions involving legal terminology, jurisdiction, procedural posture, or the authority of a source.

The practical implication is that dense retrieval is valuable for discovering conceptually related evidence,

but its results must still be assessed for legal relevance and sufficiency.

6.4 Hybrid retrieval: combining lexical and semantic evidence

Lexical and dense retrieval address complementary weaknesses. Lexical search is often effective for exact terms, identifiers, and citations; dense search can retrieve conceptually relevant passages expressed using different language. Hybrid retrieval combines these signals to reduce dependence on either one alone.

A hybrid architecture may run BM25 and dense retrieval in parallel, obtain ranked lists from both, and combine them using a fusion method. One widely used method is Reciprocal Rank Fusion (RRF), introduced by Cormack et al. (2009). RRF combines rankings using the reciprocal of each document's rank, with a constant controlling how much the rank position contributes to the final score.

Its central advantage is that it can combine rankings without requiring their original scores to be directly comparable. BM25 scores and dense similarity scores may have different distributions; rank fusion provides a way to combine their relative ordering instead. Conceptually, RRF can be written as

$$\mathrm{RRF}(d) = \sum_{r \in R} \frac{1}{k + \mathrm{rank}_r(d)}, \tag{1}$$

where d is a candidate document or passage, R is the set of ranked retrieval results, $\mathrm{rank}_r(d)$ is the rank of the candidate in retrieval list r, and k is a smoothing constant.

A passage ranked highly by multiple retrieval methods can receive a strong fused score. A passage found by only one method may still rank well if that method places it sufficiently high.

Importantly, RRF combines rankings; it does not independently verify legal correctness. If both retrieval streams return irrelevant evidence, fusion cannot turn that evidence into authoritative support. Likewise, RRF does not automatically preserve statutory hierarchy or determine whether a retrieved passage contains a decisive exception. Table 11 compares the principal retrieval strategies.

Hybrid retrieval is therefore best understood as a method for improving evidence discovery. Whether it improves final answer reliability must be tested separately through retrieval metrics and grounded-generation evaluation.

6.5 Reranking and evidence selection

Retrieval systems often return a relatively broad set of candidate passages. These candidates may be sufficient for an initial search but unsuitable for direct inclusion in a generator's context. Reranking introduces a second-stage relevance assessment that reorders candidates according to a more targeted relationship between the query and each passage.

A typical architecture first uses a relatively efficient retriever to obtain candidate passages. A reranker then evaluates the query–passage pairs more carefully and assigns a refined ranking. Only the highest-ranked evidence is passed to the generator.

In legal applications, reranking can help distinguish between a passage that merely mentions a legal concept and one that directly addresses the question. Nevertheless, relevance alone is not enough. A reranker trained to estimate textual relevance may still fail to recognize that a provision has been superseded, that a judgment is distinguishable, or that a qualification elsewhere in the statute changes the interpretation.

Evidence selection must therefore consider more than similarity. Where available, useful signals include source authority, jurisdiction, date or version, document type, and structural relationships. These signals can be incorporated into filtering, ranking, or subsequent verification, but their effectiveness depends on the quality of the underlying metadata and rules.

A further issue is context-window management. Providing every retrieved passage to the generator can introduce noise, redundancy, or conflicting information. Providing too little context can omit an exception or related provision. The goal is not simply to maximize the number of retrieved passages; it is to provide a compact and sufficiently complete set of evidence for the question.

6.6 Structure-aware indexing: preserving legal hierarchy

Legal documents are inherently hierarchical. A statute may contain parts, chapters, sections, subsections, clauses, and provisos. A judgment may contain procedural history, facts, issues, arguments, reasoning, and the final decision. These structures help determine how an individual passage should be interpreted.

Conventional chunking strategies often divide documents according to a fixed token limit or a simple text boundary. Such methods are computationally convenient, but they can separate a provision from its heading, remove the parent section from a subsection, or split an exception away from the rule it qualifies. Consider a simplified example:

> *Illustrative legal hierarchy (schematic):*
>
> - **Act**: the governing legislative document
>   - **Section 10**: general legal rule
>   - **Section 10(2)**: condition or qualification
>   - **Section 10(3)**: exception or additional requirement
>
> The example is schematic, not a statement about a particular statute.

If a system retrieves only the general rule, it may overlook a condition or exception in a subordinate provision. The retrieved text may be relevant yet insufficient to support a complete answer. Structure-aware indexing seeks to reduce this risk by preserving information about the location and relationships of passages within the original document.

Table 11: Comparison of principal retrieval strategies.

Strategy	Primary strength	Main weakness	Role in legal RAG
BM25 / lexical retrieval	Exact legal terms, names, identifiers, and statutory references.	May miss relevant passages expressed using different terminology.	Strong baseline for exact-match legal search.
Dense retrieval	Semantic similarity and paraphrase matching.	Can retrieve conceptually similar but legally inapplicable passages.	Expands retrieval beyond exact wording.
Hybrid retrieval	Combines exact-term matching and semantic search.	Requires tuning and does not guarantee evidence correctness.	Broadens the candidate evidence set.
Reciprocal Rank Fusion	Combines ranked lists without directly comparing their raw scores.	Cannot correct poor candidate lists or assess legal authority by itself.	Integrates complementary retrieval results.
Reranking	Reassesses candidate passages using a more targeted relevance model.	Adds computation and remains dependent on the quality of candidates and training.	Refines the ordering before evidence is passed to the generator.

A structure-aware index may retain hierarchical breadcrumbs, section identifiers, headings, parent-child relationships, document metadata, and source references alongside the text itself. For example, a passage can be associated with a breadcrumb such as Constitution → Part II → Chapter 1 → Article 10A. When retrieved, this information helps the system identify where the passage belongs and may support retrieval of its surrounding context.

Structure-aware indexing is not identical to semantic chunking. Semantic chunking attempts to create text segments that are coherent in meaning; structure-aware indexing additionally represents the passage's position within the source document. The two can be combined: a system may create semantically coherent chunks while retaining their statutory hierarchy and provenance.

Nor does preserving hierarchy automatically resolve legal interpretation. The system must still determine which related provisions matter to the question, whether the source is authoritative and current, and whether the combined evidence supports the conclusion. Structure-aware indexing supplies useful context; it does not replace legal reasoning or verification.

6.7 Knowledge-graph-based retrieval

Knowledge graphs offer another way to represent relationships among legal entities and authorities. Rather than storing documents only as independent passages, a graph represents entities as nodes and relationships as edges. Legal entities may include statutes, provisions, cases, courts, parties, legal concepts, and amendments. A graph can represent relationships such as:

- a provision belonging to a particular statute;
- a judgment interpreting or citing a provision;
- an amendment modifying an earlier provision;
- a court issuing a decision;
- a judgment addressing a defined legal issue.

Such relationships can support retrieval beyond direct text similarity. A system may retrieve a provision and then traverse relevant connections to locate judgments that interpret it or related provisions that constrain its application. When combined with dense or lexical retrieval, graph traversal can help connect evidence that is dispersed across documents.

However, graph quality depends on the accuracy of entity extraction, relationship identification, source linking, and temporal information. An incorrectly extracted citation can create a misleading connection. A graph that records that a judgment cites a provision does not necessarily establish that the judgment adopts, applies, or endorses the provision's interpretation.

Knowledge graphs and structure-aware indexes also solve different problems. A hierarchical index preserves a document's internal organization; a knowledge graph can represent relationships across multiple documents. They can be complementary components of a retrieval system rather than competing alternatives.

6.8 A unified retrieval architecture for legal intelligence

The preceding approaches address different parts of the retrieval problem. Lexical retrieval identifies exact textual matches; dense retrieval expands discovery to semantically related passages; rank fusion combines their results; reranking refines candidate ordering; structure-aware indexing preserves document context; and knowledge graphs represent explicit relationships among legal authorities. A unified architecture can combine these components in a staged pipeline, illustrated in Figure 1.

A knowledge graph can be connected to the context-and-authority stage to discover relationships among retrieved sources. An agentic controller may also invoke retrieval again when the evidence is incomplete or contradictory. Both additions should be justified by measurable benefits rather than included merely to increase architectural complexity.

6.9 How retrieval architectures should be evaluated

A retrieval architecture should be assessed independently from the quality of its generated answers. If the retrieval component does not return the required evidence, the generator's output cannot reliably compensate for that failure. Conversely, successful retrieval does not establish that the generator will interpret the

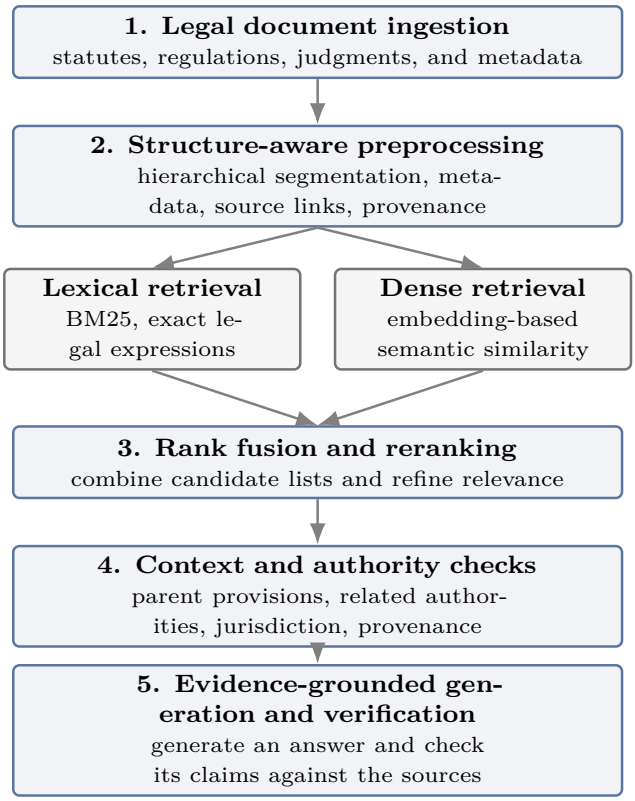


Figure 1: Conceptual architecture synthesizing complementary legal retrieval methods. This is a proposed reference architecture, not a claim that every component has been validated together in a single experiment.

evidence correctly. Several complementary metrics are useful, as summarized in Table 12.

These metrics require clear annotation guidelines. For example, relevance may be graded at the document level or the passage level; the two can produce different results. Similarly, citation correctness requires an explicit definition of what counts as support. A citation that mentions the same topic is not necessarily a citation that substantiates the claim.

A rigorous evaluation should therefore use a consistent query set and legal corpus, compare lexical, dense, hybrid, and structure-aware configurations under controlled conditions, and measure both retrieval and final answer reliability. Where feasible, ablation studies should remove one component at a time to determine whether improvements can be attributed to rank fusion, structural metadata, reranking, graph traversal, or another architectural choice.

6.10 Synthesis and remaining research challenges

The development of legal retrieval architectures reflects a progression from finding text that matches a query toward identifying evidence that is relevant, structurally contextualized, and suitable for supporting a legal conclusion. Lexical retrieval provides a strong foundation for exact legal terminology. Dense retrieval addresses differences in wording. Hybrid retrieval and rank fusion combine complementary signals, while reranking refines the candidate set. Structure-aware indexing preserves the organization of legal documents, and knowledge graphs can connect authorities across documents.

None of these components independently guarantees legal correctness. Their value depends on the legal corpus, source provenance, retrieval configuration, quality of metadata, and evaluation protocol. A sophisticated retriever can still return an incomplete set of authorities, while a well-ranked passage can remain insufficient if its parent provision or relevant exception is missing.

The resulting research direction is not simply to build a larger or more complicated retrieval pipeline. It is to design systems in which the evidence required for a legal answer can be retrieved, contextualized, and verified systematically. This provides the technical foundation for the next part of the survey, which examines hallucination, factual support, and citation correctness as separate but connected dimensions of legal AI reliability.

7 Reliability, Hallucination, and Legal Citation Verification

7.1 From relevant retrieval to reliable legal answers

Retrieval-augmented generation (RAG) has become an important approach to reducing the dependence of large language models (LLMs) on parametric knowledge alone. By retrieving external documents at inference time, a RAG system can provide answers grounded in legal statutes, regulations, judgments, and other authoritative materials. However, access to external evidence does not automatically make an answer reliable. A system may retrieve a relevant document but misinterpret its legal meaning, overlook an exception, cite the wrong passage, or generate a conclusion that the retrieved evidence does not support.

This distinction is particularly important in law because legal correctness depends on more than textual similarity. An answer may accurately reproduce a sentence from a judgment while misrepresenting its holding. It may identify the correct statute but overlook an amendment, jurisdictional restriction, procedural condition, or subsequent decision that affects its application. Consequently, reliability must be evaluated across the complete pipeline, from evidence retrieval to legal interpretation and citation generation.

The reliability problem can be understood through three related questions:

1. **Was the right evidence retrieved?** The system must locate the relevant legal provision, passage, or judgment.
2. **Does the evidence support the claim?** The generated statement must follow from the cited material rather than merely discuss a similar topic.
3. **Is the claim legally applicable?** The system must account for relevant jurisdiction, temporal validity, procedural context, and legal authority.

These questions are related but not interchangeable.

Table 12: Complementary metrics for evaluating legal retrieval architectures.

Metric	What it measures	Why it matters in legal retrieval
Recall@k	Whether relevant evidence appears among the top k retrieved results.	Measures the system's ability to find needed authorities or passages.
Precision@k	The proportion of the top k results that are relevant.	Indicates how much irrelevant material may enter the generation context.
Mean Reciprocal Rank (MRR)	The reciprocal rank of the first relevant result, averaged across queries.	Measures how early the system finds a relevant passage.
Normalized Discounted Cumulative Gain (nDCG)	Ranking quality when results have graded relevance.	Helps distinguish highly relevant legal evidence from merely related passages.
Citation precision	The proportion of supplied citations that correctly support their associated claims, under a defined annotation protocol.	Measures whether references are actually evidentially useful.
Citation recall	The proportion of required supporting sources or claims that receive appropriate citations, under a defined protocol.	Helps detect missing legal support.
Unsupported-claim rate	The proportion of evaluated claims that lack adequate support in the permitted evidence.	Evaluates grounding beyond the presence of citations.
Latency and cost	Time and computational or service cost per query.	Helps determine whether architectural improvements are practical.

A system can perform well on retrieval metrics and still generate unsupported conclusions. Similarly, a response containing accurate quotations may remain misleading if it applies them to the wrong legal issue.

7.2 Types of reliability failures

Legal AI reliability is not a single measurable property. It includes several failure modes that require different detection and mitigation strategies, as summarized in Table 13.

The categories overlap. For example, a jurisdictional mismatch may also produce an unsupported claim, while outdated law can lead to incorrect citations. Nevertheless, distinguishing the underlying failure is useful because a single safeguard cannot address every problem.

7.3 Evidence grounding and citation correctness

A citation should not be treated as reliable merely because it points to an existing document. Legal citation verification requires checking both the existence of the source and its relationship to the claim. A practical verification process separates four operations:

- **Citation resolution:** Determine whether the cited statute, case, section, paragraph, or other authority exists and can be located.
- **Passage verification:** Confirm that the cited passage contains the relevant legal language or reasoning.
- **Claim-support verification:** Assess whether the passage actually supports the claim, rather than simply sharing vocabulary with it.
- **Applicability verification:** Check whether the authority is relevant to the jurisdiction, date, issue, and procedural context of the question.

This process suggests that citation verification should operate at the level of individual claims, not merely at the level of the complete response. A long answer may contain several supported statements and one critical unsupported conclusion. Evaluating the answer as a single unit can obscure that distinction.

LegalBench-RAG addresses an important part of this problem by focusing on retrieving precise legal text segments rather than relying solely on document-level retrieval. Its benchmark design reinforces the importance of locating evidence at a sufficiently granular level for downstream legal tasks (Pipitone and Houir Alami, 2024). However, precise retrieval and legal entailment remain separate requirements: finding the correct passage does not itself establish that the generated interpretation is valid.

Dahl et al. (2024), in their discussion of fabricated legal authorities and the broader reliability of generative AI in law, and Magesh et al. (2025), in their assessment of leading AI legal research tools, further motivate evaluating the output rather than assuming that fluent language or a citation list indicates correctness. These studies should not be interpreted as establishing one universal hallucination rate for all legal AI systems. Results depend on the tools, tasks, jurisdictions, test sets, and definitions of error used in each evaluation.

7.4 Evaluating reliability: beyond answer accuracy

A robust evaluation framework should distinguish retrieval quality, citation quality, factual or legal support, and operational performance. No single metric captures all of these dimensions, as Table 14 shows.

Metric definitions should be fixed before evaluation. For example, citation validity rate and citation support rate should not be used interchangeably. A citation can be valid in the first sense and invalid in the second. Likewise, an unsupported-claim rate should specify whether the denominator consists of all generated claims, all material claims, or all claims requiring citations.

Evaluation should also distinguish automatic checks

Table 13: Reliability failure modes in legal AI and possible mitigations.

Failure type	**Description**	**Illustrative example**	**Possible mitigation**
Fabricated authority	The system invents a case, statute, provision, or citation.	Referring to a nonexistent judgment.	Citation resolution against authoritative databases.
Incorrect citation	A real source is cited, but the cited passage does not support the associated claim.	Citing a procedural provision as authority for a substantive conclusion.	Claim-to-passage entailment checking.
Unsupported claim	The response makes a factual or legal assertion that is not established by retrieved evidence.	Stating that a remedy is available without retrieving the relevant conditions.	Evidence coverage checks and abstention.
Misinterpretation	The source is real, but its legal meaning is misrepresented.	Treating an observation as the court's binding holding.	Role-aware judgment analysis and contextual verification.
Incomplete evidence	A relevant exception, counter-authority, or condition is omitted.	Explaining a general rule without its statutory exception.	Query expansion and retrieval of related provisions.
Outdated law	The system relies on a superseded or inapplicable version of a legal source.	Applying an earlier statutory amendment.	Effective-date metadata and version-aware retrieval.
Jurisdictional mismatch	The system applies law from a different jurisdiction.	Using an Indian legal rule to answer a Pakistani-law question.	Jurisdiction filtering and authority classification.
Contradictory evidence	The system fails to reconcile conflicting authorities.	Presenting one judgment without acknowledging a relevant later ruling.	Authority-aware ranking and explicit conflict reporting.

Table 14: Dimensions for evaluating legal AI reliability.

Evaluation dimension	**Example metric**	**What it measures**	**Important limitation**
Retrieval effectiveness	Recall@k	Whether relevant evidence appears among the top k retrieved results.	Does not measure whether the final answer uses that evidence correctly.
Ranking quality	MRR or nDCG	How effectively the system ranks relevant passages.	Depends on relevance judgments and benchmark design.
Citation validity	Citation validity rate	Proportion of generated citations that resolve to real sources.	A real citation may still be irrelevant or misleading.
Citation support	Claim–citation support rate	Proportion of claims adequately supported by their cited evidence.	Requires careful annotation or reliable support verification.
Evidence coverage	Evidence coverage rate	Extent to which material claims have supporting evidence.	Does not establish that the evidence is legally correct or complete.
Unsupported content	Unsupported-claim rate	Proportion of claims lacking adequate evidential support.	Results vary with claim segmentation and annotation policy.
Legal correctness	Expert-assessed correctness	Whether the answer's legal analysis is judged correct under the task conditions.	Requires qualified evaluators and explicit jurisdictional criteria.
Calibration	Confidence–accuracy alignment	Whether confidence corresponds to observed correctness.	Model-reported confidence can be poorly calibrated.
Efficiency	Latency and cost per query	Operational expense of producing an answer.	Low cost or latency does not imply legal reliability.

from expert assessment. Automated tools can identify broken links, malformed citations, missing sources, and some forms of semantic inconsistency. Human legal reviewers remain important for assessing nuanced statutory interpretation, the relative authority of cases, exceptions, and whether a response appropriately qualifies uncertainty. Ideally, evaluations should report the annotation protocol, reviewer expertise, agreement between reviewers, and adjudication procedure.

7.5 Safeguards for legal AI applications

Reliability is best addressed through multiple safeguards rather than a single verification component:

1. **Authoritative source management.** Record provenance, jurisdiction, document identity, effective dates, and amendments.
2. **Evidence retrieval and ranking.** Retrieve relevant passages using lexical, dense, hybrid, or structure-aware methods.
3. **Claim-level generation.** Associate material legal claims with specific supporting passages.
4. **Verification and conflict checks.** Resolve citations, test claim support, and identify missing or conflicting authorities.
5. **Qualified response or abstention.** Answer with traceable evidence, state uncertainty, or decline to make an unsupported conclusion.

This is a conceptual reliability workflow synthesized for this survey; it is not a claim that every reviewed system implements all five stages.

The workflow highlights several design principles. First, provenance and temporal metadata should be preserved during ingestion rather than added after generation. Second, verification should be applied to material claims and their supporting passages. Third, the system should be allowed to abstain when evidence is missing or contradictory. Finally, high-impact legal applications should preserve access to the underlying sources and support review by a qualified human.

Agentic architectures can coordinate these safeguards through retrieval, verification, and revision steps. However, adding more agents does not automatically improve correctness. If the agents rely on the same incomplete corpus or repeat the same unsupported assumption, orchestration may reproduce rather than correct the error. Reliability therefore depends on independent evidence checks, explicit verification criteria, and measurable performance improvements.

7.6 Section synthesis

The central lesson is that legal AI reliability must be evaluated as an evidence-to-claim problem. Retrieval quality is necessary, but it is not sufficient. A dependable system must retrieve appropriate authority, preserve its context, generate claims that follow from the evidence, verify citations, and recognize when legal applicability remains uncertain. These requirements motivate the comparative analysis in the next section, where contemporary systems are assessed according to their evidence sources, retrieval design, legal reasoning, verification mechanisms, and evaluation practices.

7.7 Implications for evaluation and reproducibility

The literature also highlights a methodological problem: reliability results cannot be compared fairly unless the underlying evaluation conditions are reported. A hallucination rate obtained from open-ended questions about legal doctrine is not directly comparable with one obtained from citation verification, statutory question answering, or retrieval of annotated passages.

For example, Magesh et al. (2025) evaluated commercial legal research tools and found that hallucinations persisted despite the use of RAG. Their findings challenge claims that retrieval alone can guarantee hallucination-free outputs. Dahl et al. (2024) likewise documented substantial hallucination problems in general-purpose LLMs answering legal questions. A reproducible legal AI evaluation should therefore disclose:

- the jurisdiction, legal domain, and temporal scope of the corpus;
- the source of questions and the criteria used to construct the test set;
- the operational definition of hallucination and unsupported claims;
- the annotation instructions, reviewer expertise, and agreement statistics;
- the retrieval configuration, model version, prompts, and verification settings;
- the baseline systems, uncertainty estimates, and statistical tests where appropriate.

These requirements are especially important when evaluating improvements from hybrid retrieval, reranking, knowledge graphs, or agentic verification. Each component should be assessed through controlled comparisons or ablation studies, rather than attributing all observed improvements to the complete architecture.

The resulting research priority is clear: legal AI needs not only more capable models, but also transparent evidence pipelines and evaluations that measure whether answers are actually supported by applicable legal authority.

8 Comparative Analysis of Contemporary Legal AI Systems

8.1 A common framework for comparing legal AI architectures

Contemporary legal AI systems differ not only in the models they use but also in how they retrieve, organize, interpret, and verify legal evidence. A fair comparison must therefore examine the complete processing pipeline rather than focusing exclusively on the underlying language model or a reported accuracy score.

This section compares representative systems using six dimensions:

1. **Architecture:** how the system organizes its processing components.
2. **Retrieval strategy:** how it identifies and ranks relevant legal material.
3. **Generation method:** how it produces classifications, legal analysis, or drafted text.
4. **Evidence verification:** whether it checks source validity and claim-level support.
5. **Evaluation:** how its performance is measured.
6. **Limitations:** what its architecture or experimental design does not establish.

The comparison distinguishes between legal language models, benchmarks, retrieval systems, and end-to-end applications. This distinction is necessary because a benchmark such as LegalBench-RAG evaluates a specific capability, whereas a platform such as JusBuild is designed to support a legal workflow. Their contributions are complementary, but they cannot be ranked directly without common experiments. A benchmark for legal classification cannot establish that a system is reliable at retrieving current statutes, and a document-drafting system cannot automatically be considered superior to a legal search system.

8.2 Architecture and retrieval strategy

Table 15 compares the architecture and retrieval strategy of representative legal AI approaches.

The table demonstrates that architectural differences correspond to different research objectives. Domain-adapted transformers aim to improve the model's understanding of legal language. Retrieval benchmarks measure whether relevant evidence can be found. End-to-end RAG applications combine retrieval with generation, while structure-aware systems additionally attempt to preserve relationships within legal documents.

These differences matter because a model that performs well on a legal classification task may not retrieve the relevant statutory provision, and a system that retrieves the right passage may still misinterpret its legal meaning. The architecture must therefore be assessed against the particular failure modes it is intended to address.

8.3 Evidence verification and evaluation

Table 16 summarizes evidence handling and evaluation considerations across the same systems.

A critical distinction emerges: citation presence, citation validity, and citation support are three different properties. A system may generate a citation, point to an existing legal source, and still misrepresent what that source establishes. Accordingly, evaluation should measure whether citations resolve correctly and whether the cited passages support individual claims.

The same caution applies to numerical results. Retrieval recall, legal answer accuracy, unsupported-claim rate, and citation precision measure different properties. Scores reported on different datasets or under different annotation protocols should not be compared as if they were measurements of the same outcome. For this reason, contemporary systems should ideally be evaluated through a common protocol combining passage-level retrieval metrics, expert-assessed legal correctness, citation-support analysis, temporal and jurisdictional checks, and operational measures such as latency and cost.

8.4 Comparing system design philosophies

The systems above illustrate three broad design philosophies.

8.4.1 Model-centric legal AI

Model-centric approaches improve legal performance by adapting language models to legal text, tasks, or reasoning patterns. LEGAL-BERT and research evaluated using LexGLUE exemplify this direction. LegalBench complements it by testing a wider range of legal reasoning capabilities.

The principal advantage is that domain adaptation can improve a model's ability to interpret legal terminology, identify relevant concepts, and perform specialized NLP tasks. However, model-centric systems remain limited when a question depends on an authoritative source that was absent from training, has changed since training, or must be cited precisely.

The key distinction is between understanding legal language and retrieving legally applicable evidence. A model can recognize the meaning of a statutory provision without establishing whether that provision is current or whether it applies to the user's specific question.

8.4.2 Retrieval-centric legal AI

Retrieval-centric approaches place greater emphasis on locating relevant legal material. LegalBench-RAG demonstrates the importance of retrieving precise text segments rather than returning only broad document-level matches. The Pakistani legal RAG framework illustrates the use of hybrid retrieval to combine semantic similarity with keyword matching.

The main advantage is that answers can be conditioned on external legal sources, reducing dependence on the model's internal knowledge. Retrieval can also make responses more auditable because a user can inspect the source passages.

Nevertheless, retrieval quality depends on corpus coverage, document parsing, chunking, query interpretation, and ranking. Even a highly relevant passage can be misinterpreted or applied outside its proper legal context. Retrieval-centric systems therefore need a separate mechanism for validating the relationship between evidence and generated claims.

Table 15: Architectural comparison of representative legal AI approaches.

System	Architecture	Retrieval strategy	Generation or output method
LEGAL-BERT (Chalkidis et al., 2020)	Legal-domain pretrained transformer	Not inherently a retrieval system	Contextual representations for downstream legal NLP tasks
LexGLUE (Chalkidis et al., 2022)	Benchmark suite for legal language understanding	Depends on the evaluated model and task	Task-specific predictions, such as classification
LegalBench (Guha et al., 2023)	Multi-task legal reasoning benchmark	Not a single retrieval architecture	Model-generated answers or task outputs evaluated against task-specific criteria
LegalBench-RAG (Pipitone and Houir Alami, 2024)	Retrieval benchmark with annotated legal text segments	Evaluates retrieval of relevant legal passages	Evaluates retrieval rather than prescribing one answer-generation architecture
JusBuild (Buffa et al., 2025)	Structured legal document-building workflow	Document segmentation and vector-based retrieval	Combines retrieved precedent material with AI-assisted drafting and human oversight
Pakistani legal RAG framework (Ali et al., 2025)	Jurisdiction-specific question-answering pipeline	Hybrid keyword and semantic retrieval	LLM-generated legal answers grounded in retrieved materials
Ren et al. (Ren et al., 2026)	Legal LLM improvement pipeline incorporating data synthesis and RAG	Retrieval-augmented generation alongside synthetic-data development	Legal LLM outputs assessed through benchmark and model-evaluation procedures
Legalyze (Hamza et al., 2026)	Structure-aware, agentic legal RAG platform	Hierarchical indexing, lexical and dense retrieval, reciprocal-rank fusion	Agentic legal analysis and synthesis grounded in retrieved legal evidence

Table 16: Evidence handling and evaluation considerations.

System or approach	Evidence verification	Evaluation focus	Key limitation to consider
LEGAL-BERT	No inherent citation-verification mechanism	Performance on downstream legal NLP tasks	Legal language competence does not establish evidence-grounded reasoning
LexGLUE	Determined by the evaluated task and model	Standardized legal language understanding	Task scores do not directly measure citation support
LegalBench	Task-dependent; not a universal citation verifier	A broad range of legal reasoning tasks	Performance may vary considerably by task and legal concept
LegalBench-RAG	Evaluates retrieval against annotated relevant passages	Passage-level retrieval quality	Does not, by itself, establish the correctness of a generated legal answer
JusBuild	Uses a structured drafting workflow with human oversight	Legal document-building performance in its evaluated settings	Human review and drafting performance do not establish general-purpose legal reliability
Pakistani legal RAG framework	Evidence is retrieved for grounded answering; verification should be assessed separately	Legal question answering over selected Pakistani legal materials	Grounding alone does not establish source completeness, temporal validity, or claim-level entailment
Ren et al.	Retrieval and evaluation components support the broader legal LLM pipeline	Legal model performance and evaluation of generated outputs	Results depend on the selected benchmarks and experimental settings
Legalyze	Citation-grounded response design and a hybrid retrieval pipeline	Retrieval and answer reliability within its reported evaluation setting	Performance on one corpus and query set cannot establish universal legal correctness

8.4.3 Structure-aware and workflow-oriented legal AI

A third direction combines retrieval with legal document structure, task-specific processing, or coordinated workflows. JusBuild uses document segmentation and section-level retrieval to assist with structured drafting. Legalyze combines hierarchical legal indexing with lexical and dense retrieval, rank fusion, and agentic analysis.

These approaches recognize that legal documents are not simply collections of independent text passages. A provision belongs to a section, chapter, statute, or legal instrument; a judgment contains procedural history, arguments, reasoning, and a decision. Preserving this organization can help a system retrieve the correct passage while retaining the context needed to interpret it.

Workflow-oriented designs can additionally separate retrieval, analysis, synthesis, and verification. Their potential benefit is improved traceability and more explicit control over intermediate operations. Their cost is architectural complexity: every additional stage introduces implementation, latency, and evaluation requirements. The critical question is not whether a system has more components, but whether those components address identifiable failure modes and produce measurable improvements.

8.5 Cross-system comparison: capabilities and unresolved gaps

The comparison can be extended beyond individual systems to the capabilities that a dependable legal AI platform should support, as set out in Table 17.

This comparison reveals that the main challenge is not necessarily the absence of an individual technique. Rather, the challenge is integrating these techniques into a coherent pipeline while preserving evidence provenance and enabling meaningful evaluation. For example, hybrid retrieval and reranking can improve candidate selection, but neither guarantees that the selected passage is legally authoritative. Hierarchical indexing can preserve document context, but it does not automatically establish relationships between separate judgments and statutes. A knowledge graph can encode those relationships, but its usefulness depends on the accuracy, completeness, and provenance of the graph. Citation verification can identify unsupported claims, but only if the verification process has access to suitable evidence and appropriate legal criteria.

8.6 The Pakistani legal AI context

The Pakistani legal context provides a useful case study because jurisdiction-specific legal materials must be interpreted within their own institutional and documentary structures. A system developed primarily around US or European legal resources cannot be assumed to perform reliably on Pakistani statutes and judgments without appropriate corpus construction, jurisdictional adaptation, and evaluation.

Ali et al. (2025) describe a RAG-based framework for Pakistani legal question answering that combines semantic and keyword-based retrieval over selected legal materials, including the Code of Criminal Procedure and Pakistan Penal Code. This work illustrates the importance of combining legal-domain evidence with retrieval methods suited to the source material.

Legalyze (Hamza et al., 2026) follows a related but more structure-aware approach. Its architecture incorporates hierarchical indexing, lexical and dense retrieval, reciprocal-rank fusion, and agentic legal analysis over Pakistani legal materials. These design choices address complementary challenges: lexical retrieval supports exact legal terminology, dense retrieval supports semantic variation, rank fusion combines result lists, and hierarchical indexing helps preserve legal document organization.

However, a meaningful comparison between the two systems would require a shared evaluation dataset, consistent query sets, documented corpus coverage, equivalent retrieval budgets, and common measures of retrieval and answer quality. The existence of hybrid retrieval in both systems is not sufficient to establish which performs better. Likewise, differences in reported hallucination rates should not be attributed to a single architectural component unless the experimental design isolates that component.

A further research opportunity concerns multilingual access. Pakistani legal materials and user questions may involve English, Urdu, and Roman Urdu, while statutory text and case-law terminology may not map neatly across languages. A robust system must therefore align the user's intended legal concept with the authoritative terminology used in the source corpus, rather than relying exclusively on literal translation.

8.7 A unified framework for comparing legal AI systems

The preceding analysis suggests that future comparisons should separate four levels of system performance:

1. **Retrieval performance.** Does the system find the relevant legal material and rank it appropriately? *Measures:* Recall@k, MRR, nDCG, passage-level precision.

2. **Evidence integrity.** Does the system preserve source identity, document context, temporal validity, and jurisdiction? *Measures:* source resolution, provenance completeness, temporal and jurisdictional error rates.

3. **Legal answer quality.** Does the answer correctly interpret and apply the retrieved authority? *Measures:* expert-assessed correctness, claim support, citation accuracy, unsupported-claim rate.

4. **Operational performance.** Can the system provide useful results within acceptable computational and review costs? *Measures:* latency, computational cost, failure recovery, and human review burden.

This framework discourages the use of one aggregate score to conceal important weaknesses. A system may

Table 17: Capabilities of a dependable legal AI platform and associated research directions.

Capability	Why it matters	Typical weakness if omitted	Research direction
Legal language understanding	Interprets legal terminology and complex syntax.	Relevant legal meaning may be misunderstood.	Legal-domain adaptation and task-specific evaluation.
Hybrid retrieval	Finds both exact legal expressions and semantically related passages.	Statutory identifiers or paraphrased queries may be missed.	Combine lexical and dense retrieval.
Rank fusion and reranking	Improves the ordering of candidate evidence.	Relevant passages may remain buried in the result list.	Evaluate fusion and reranking through ablations.
Hierarchical indexing	Preserves relationships between passages and their parent documents.	Retrieved fragments may lose statutory or judgment context.	Structure-aware segmentation and metadata.
Knowledge-graph retrieval	Represents explicit relations between legal entities and authorities.	Cross-document relationships may remain difficult to retrieve.	Graph construction, entity linking, and graph–text retrieval.
Citation verification	Tests whether cited sources exist and support generated claims.	Responses may contain invalid or misleading citations.	Claim-level evidence verification.
Temporal and jurisdictional control	Helps ensure that retrieved law is applicable.	Superseded or foreign legal authorities may be used.	Versioned corpora and jurisdiction-aware filtering.
Multilingual retrieval	Supports queries expressed in different languages or scripts.	Users may fail to retrieve evidence phrased differently from their query.	Cross-lingual retrieval, transliteration, and legal terminology alignment.
Reproducible evaluation	Makes performance claims independently testable.	Improvements may reflect dataset or evaluation differences.	Public test sets, transparent protocols, and statistical analysis.

retrieve highly relevant passages but generate inaccurate legal conclusions; another may generate convincing answers but rely on incomplete or outdated evidence. Reporting each level separately makes such trade-offs visible. Future comparative studies should also report whether the system is open-source, whether its corpus and evaluation data are accessible, whether its outputs can be reproduced, and whether its architecture supports independent component replacement.

8.8 What the comparison reveals

Three conclusions follow from the comparison. First, no single architectural family addresses every legal AI problem. Domain-specific language models improve legal language processing, retrieval systems improve access to external evidence, and structured workflows can coordinate more specialized operations.

Second, evidence verification remains distinct from evidence retrieval. Lexical retrieval, dense retrieval, reranking, and rank fusion can improve the candidate evidence presented to a model. They do not automatically establish that the final answer is supported by that evidence.

Third, jurisdiction and document structure should be treated as architectural requirements rather than optional metadata. A system that retrieves semantically similar passages without considering the governing jurisdiction, the document hierarchy, or the effective date of a provision can return plausible but legally inapplicable evidence. These observations motivate the next section, which examines jurisdiction-specific reasoning, legal hierarchy, and multilingual access.

9 Jurisdiction, Document Structure, and Multilingual Legal AI

9.1 Why legal information retrieval is jurisdiction-dependent

Legal information retrieval differs from general-purpose information retrieval because the relevance of a document depends not only on its subject matter but also on its legal authority, jurisdiction, temporal validity, and relationship to the question being asked. Two passages may discuss the same legal principle while having different implications because they originate from different jurisdictions or occupy different positions within a legal system.

For example, a question concerning criminal procedure in Pakistan should prioritize applicable Pakistani legislation and relevant Pakistani judicial decisions. A semantically similar provision from another jurisdiction may help explain a general concept, but it should not automatically be treated as governing authority. Jurisdiction-sensitive retrieval must therefore consider several factors:

- **Territorial and institutional jurisdiction:** whether the authority applies to the relevant court, territory, or legal institution.
- **Authority type:** whether a document is legislation, a regulation, a binding judgment, a persuasive judgment, or secondary commentary.
- **Temporal validity:** whether the provision or interpretation was applicable at the relevant time.
- **Procedural context:** whether the authority addresses the procedural stage and legal issue involved.
- **Precedential relationships:** whether a judgment

follows, distinguishes, overrules, or otherwise relates to another decision.

These considerations expose a limitation of conventional semantic retrieval. An embedding model may identify two passages as highly similar without determining whether either passage is legally applicable. Consequently, jurisdictional and temporal constraints should influence retrieval and verification, not merely appear as explanatory text in the final answer.

9.2 Legal hierarchy and document structure

Legal documents contain multiple layers of organization. Statutes may be divided into parts, chapters, sections, subsections, clauses, and provisos. Judgments may contain procedural histories, questions of law, arguments, judicial reasoning, and final orders. These structural elements help determine the meaning and applicability of individual passages.

A conventional chunking strategy often divides documents into fixed-size text segments or overlapping windows. Although computationally convenient, this can separate a provision from its heading, definition, exception, proviso, or parent section. The resulting fragment may remain semantically understandable while losing context that is essential to legal interpretation. Consider a hypothetical example:

Illustrative statutory structure (schematic):

- **Chapter III: General Conditions**
  - **Section 12 (general rule):** a person may exercise the stated right subject to the conditions specified in this section.
    - **Subsection (2) (limitation):** the right is subject to the additional restrictions specified in the subsection.
    - **Proviso:** an exception applies where the specified conditions are satisfied.

This is an illustrative structure only, not an actual statutory provision.

If retrieval returns only the general rule, the generated answer may omit the limitation or exception. If it returns only the proviso, the system may fail to explain the general rule to which that exception applies. Structure-aware retrieval addresses this problem by preserving the relationship between a passage and its parent document elements. Useful techniques include:

1. **Hierarchical metadata:** storing the document, chapter, section, subsection, and paragraph identifiers associated with each passage.
2. **Context-preserving chunking:** retaining headings, definitions, exceptions, and relevant neighboring passages.
3. **Parent-child retrieval:** retrieving a focused passage while allowing the system to recover its broader context.
4. **Reference resolution:** linking references to other sections, definitions, schedules, and related legal instruments.
5. **Provenance preservation:** maintaining the source document, version, location, and other information needed to verify the passage.

Structure-aware indexing and knowledge-graph retrieval are complementary. Hierarchical indexing preserves the organization of a document, whereas a knowledge graph represents explicit relationships between entities, legal provisions, judgments, and other objects. A system may use both, but each requires separate construction and evaluation.

9.3 Pakistani legal information retrieval

Pakistan provides a useful case study of the challenges involved in building jurisdiction-aware legal AI. Its legal information needs span constitutional provisions, statutes, procedural codes, judicial decisions, and other legal materials. A useful system must preserve the relationships among these sources rather than treating every retrieved passage as an interchangeable text fragment.

For example, the Constitution of Pakistan, the Pakistan Penal Code (PPC), the Code of Criminal Procedure (CrPC), the Qanun-e-Shahadat Order (QSO), and judicial decisions serve different legal functions. The appropriate source depends on the question. A constitutional question may require a constitutional provision and relevant case law, while a question about a criminal procedure may require the applicable procedural provision and related judicial interpretation.

Three challenges are particularly important. First, *document coverage and authority.* A corpus may contain a large volume of legal text but still omit relevant amendments, judgments, or authoritative documents. Corpus size alone is therefore an inadequate measure of legal coverage. Second, *legal relationships.* A statutory provision may depend on a definition elsewhere in the same enactment or be interpreted by a later judgment. Retrieval should preserve these relationships where relevant, rather than relying exclusively on textual similarity. Third, *temporal and jurisdictional applicability.* A provision's wording, legal status, or interpretation may change over time. Systems should record source dates and applicable versions, and distinguish the date of a judgment from the period in which the underlying law applied.

A jurisdiction-aware Pakistani legal retrieval system should consequently combine source provenance, hierarchical indexing, authority-aware ranking, and temporal metadata. These features can improve the reliability of evidence selection, although they do not eliminate the need for legal interpretation and claim-level verification.

9.4 English, Urdu, and low-resource legal retrieval

Multilingual legal AI introduces an additional challenge: the language of a user's query may differ from the language, terminology, or script of the authoritative source. This is particularly relevant when users express legal questions in Urdu or Roman Urdu while the source material they need is written in English.

A multilingual system must bridge differences in vocabulary, spelling, transliteration, morphology, and legal terminology. Direct translation alone may be insufficient because legal concepts can have jurisdiction-specific meanings that do not map exactly across languages. Four technical approaches are particularly relevant, as summarized in Table 18.

A practical architecture can combine these methods rather than relying on one. Language identification and transliteration normalization can first process the query. Legal entity linking and terminology alignment can then identify important statutes, provisions, courts, or legal concepts. Retrieval can search both the original query and its normalized or translated variants, followed by rank fusion and legal-context filtering.

The system should preserve the original user query and the original authoritative passage. Translation can support retrieval, but the final answer should not silently substitute a translated interpretation for the governing source text.

Evaluation should also be multilingual. An English-only benchmark cannot establish whether the system works reliably for Urdu or Roman Urdu. Separate evaluation should measure retrieval recall, citation support, legal correctness, and robustness to spelling and transliteration variation across each supported query language.

9.5 Integrating jurisdiction, hierarchy, and multilingual processing

The three challenges discussed above are interdependent. Language normalization may help locate a relevant provision, but document structure is needed to preserve its context. Hierarchical indexing may identify the correct section, but jurisdictional and temporal checks are still required to determine its applicability. A conceptual architecture connecting these stages is shown in Figure 2.

This architecture offers a research framework for connecting multilingual query understanding, legal document structure, knowledge graphs, retrieval, and evidence verification. It also provides a basis for modular implementation: individual components can be replaced or evaluated independently without redesigning the entire platform.

9.6 Section synthesis

Jurisdiction, document structure, and language are not peripheral concerns in legal information retrieval. They determine which evidence should be retrieved, how its meaning should be interpreted, and whether it applies to the question being asked.

For Pakistani legal AI, the main implication is that effective retrieval requires more than semantic similarity. Systems should preserve legal hierarchy, authority relationships, temporal metadata, source provenance, and cross-language concept alignment. These requirements lead directly to the open challenges discussed next.

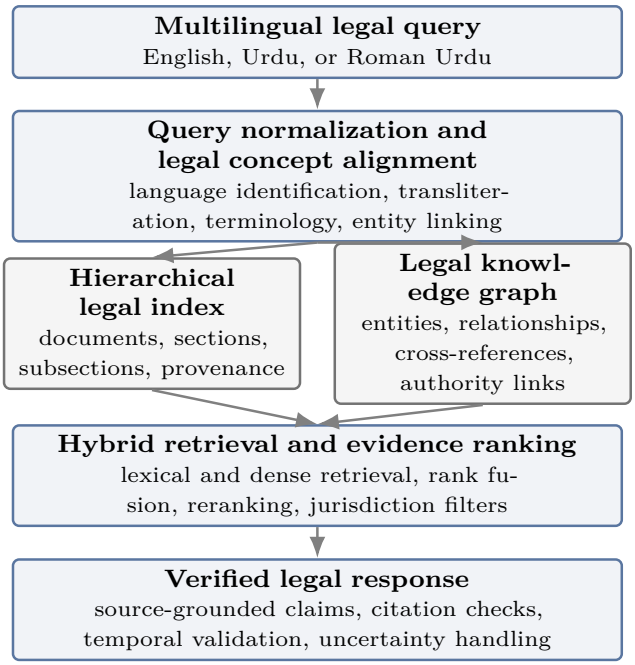


Figure 2: Conceptual architecture connecting multilingual query understanding, legal document structure, knowledge graphs, retrieval, and evidence verification. Synthesized for this survey; its individual components require empirical evaluation before any performance improvement can be claimed.

10 Open Challenges and Future Research Directions

10.1 From capable legal AI to dependable legal information retrieval

The preceding sections show that contemporary legal AI has made progress in legal language understanding, evidence retrieval, document drafting, and agentic processing. However, these capabilities remain difficult to integrate into a system that consistently returns relevant, applicable, and verifiable legal answers.

The central research challenge is therefore not simply to produce more fluent legal text. It is to develop systems that can retrieve authoritative evidence, preserve its legal context, reason over relationships between sources, and verify the claims they generate. The following research directions emerge from the comparison of legal AI architectures and the challenges of jurisdiction-specific retrieval.

10.2 Research direction 1: Knowledge-graph-grounded legal retrieval

Conventional retrieval methods primarily rank text passages according to lexical or semantic relevance. While effective for many questions, they may struggle with queries requiring relationships across multiple documents. For example, answering a complex legal question may require connecting a statutory provision to a definition, identifying judgments that interpret it, and determining whether a later authority modifies or distinguishes an earlier interpretation. A flat in-

Table 18: Technical approaches to multilingual legal retrieval.

Approach	How it works	Main advantage	Main limitation
Query translation	Translates the query into the language of the source corpus.	Enables use of existing monolingual retrieval infrastructure.	Translation may distort legal terms or procedural meaning.
Multilingual dense retrieval	Maps queries and documents from different languages into a shared embedding space.	Supports cross-language semantic matching.	Similar embeddings do not guarantee legal-concept equivalence.
Transliteration normalization	Normalizes alternative representations of words across scripts.	Helps retrieve Roman Urdu and Urdu variants.	Many transliterations are ambiguous or inconsistent.
Legal terminology alignment	Links expressions to jurisdiction-specific legal concepts and authoritative terms.	Improves conceptual consistency.	Requires curated terminology, entity linking, or legal-domain resources.

dex may retrieve individual passages without explicitly representing these relationships.

Knowledge graphs provide a complementary representation by encoding entities and relationships as structured edges. In legal applications, graph entities might represent statutes, provisions, judgments, courts, parties, legal concepts, and dates. Relationships could represent references, interpretation, amendment, procedural connections, or other explicitly supported links.

Future research should investigate how to combine graph-based retrieval with lexical and dense retrieval while preserving provenance for every relationship. The graph must distinguish explicitly stated facts from inferred connections and should not encode a legal relationship merely because two documents discuss similar subjects. Key research questions include:

- Does graph-grounded retrieval improve recall for multi-hop legal questions compared with hybrid text retrieval alone?
- Can graph relationships improve the identification of relevant authorities and exceptions?
- How should uncertain, conflicting, or incomplete graph relationships be represented?
- What methods can update the graph when statutes are amended or new judgments become available?

The most useful evaluation would compare a text-only baseline, a hybrid retrieval system, a graph-only method, and a combined graph–text architecture using the same test queries and evidence annotations.

10.3 Research direction 2: Multilingual and low-resource legal retrieval

Multilingual legal retrieval remains challenging when queries are expressed in languages or scripts that are poorly represented in training data. English, Urdu, and Roman Urdu may express similar legal intentions through different lexical forms, transliterations, or terminology.

Future research should investigate cross-lingual retrieval models, legal terminology alignment, transliteration-aware indexing, and query expansion. Rather than translating every query into English and discarding the original wording, a system could search across multiple query representations and fuse the results. The principal challenge is to distinguish linguistic similarity from legal equivalence. A translated phrase may be grammatically correct but fail to represent a jurisdiction-specific legal concept accurately. Useful research questions include:

- How does retrieval performance vary between English, Urdu, and Roman Urdu queries?
- Does multilingual dense retrieval outperform translation-only and transliteration-only baselines?
- Can legal entity linking reduce errors involving statutory names, section numbers, courts, and legal terminology?
- How can multilingual benchmarks evaluate both retrieval relevance and legal correctness?

A reproducible benchmark should include queries in each supported language, authoritative source passages, equivalent or carefully aligned legal intents, and explicit judgments of whether retrieved evidence is applicable. It should also test spelling variation, code-switching, and unseen transliteration forms.

10.4 Research direction 3: Structure-aware legal indexing

Legal document structure is frequently lost during conversion, chunking, and indexing. This creates a risk that the retrieval system will return a fragment without the definitions, exceptions, headings, or parent sections needed to interpret it correctly.

Structure-aware indexing offers a way to preserve these relationships. Future research should compare fixed-size chunking, semantic chunking, hierarchical indexing, and parent-child retrieval under the same evaluation conditions. A particularly important question is whether preserving structure improves downstream answer quality beyond its effect on retrieval recall. For example, a structure-aware system may retrieve a relevant subsection and its parent heading, allowing the generator to interpret the provision more accurately even if the retrieved subsection itself is unchanged.

Research should also examine how to handle mal-

formed PDFs, inconsistent numbering, footnotes, tables, nested clauses, and cross-references. These are not merely preprocessing inconveniences: errors during document parsing can propagate through retrieval and generation. Promising evaluation measures include:

- structural extraction accuracy for headings, sections, and subsections;
- retrieval recall for provisions whose meaning depends on parent context;
- accuracy of cross-reference resolution;
- citation localization accuracy at the section or paragraph level;
- downstream legal correctness under controlled comparisons.

10.5 Research direction 4: Claim-level evidence verification

Citation-grounded generation is insufficient if the system only checks whether a response contains references. Future legal AI architectures should explicitly associate each material claim with the passage or passages intended to support it.

A claim-level verification pipeline could identify factual and legal assertions, resolve their citations, compare each claim with its evidence, and flag unsupported or contradictory statements. Depending on the application, verification might combine deterministic citation resolution, metadata checks, textual entailment models, and qualified human review. The research challenge is that legal support is often more complex than ordinary textual entailment. A passage may establish a general rule while a different passage provides an exception. A judgment may discuss an argument without adopting it. A later authority may change how an earlier decision should be understood.

Future work should therefore evaluate more than whether a cited passage is topically relevant. It should assess whether the passage supports the claim in context and whether material qualifications or counter-authorities have been omitted. Possible research questions include:

- How accurately can automated methods distinguish supported, contradicted, and unsupported legal claims?
- Can verification identify citations to the wrong subsection or an irrelevant paragraph within an otherwise valid authority?
- Does claim-level verification reduce unsupported-claim rates without introducing excessive false alarms?
- When should the system abstain rather than generate a legal conclusion?

An effective benchmark should include claim-level annotations, source locations, support labels, and cases involving exceptions or conflicting authorities.

10.6 Research direction 5: Agentic legal AI with explicit quality gates

Agentic AI can coordinate multiple operations, such as query decomposition, iterative retrieval, legal analysis, and citation checking. However, orchestration introduces additional opportunities for error. An agent may select an inappropriate tool, carry forward a mistaken assumption, or produce a confident conclusion based on incomplete evidence.

Future research should focus on evidence-controlled orchestration rather than simply increasing the number of agents. Each stage should have explicit input requirements, output contracts, and quality checks. For example, an agent responsible for legal analysis should receive retrieved evidence with source identifiers and relevant document context. A verification stage should test whether the proposed claims are supported before the final response is released. If evidence is missing, the system should trigger additional retrieval or qualify its answer.

Software engineering principles are particularly relevant here. Modular interfaces, replaceable components, explicit error handling, versioned configurations, and regression tests can make agentic legal systems easier to maintain and evaluate. Potential experiments should compare a single-agent baseline with multi-stage and multi-agent alternatives under equivalent retrieval and generation settings. Measurements should include legal correctness, citation support, failure recovery, latency, computational cost, and the rate of unsupported claims. Without such controls, improvements cannot reliably be attributed to agentic orchestration.

10.7 Research direction 6: Temporal and authority-aware reasoning

Legal information changes over time. Statutes may be amended, judgments may be distinguished or overruled, and procedural requirements may change. A retrieval system that ignores these changes can produce an answer that was once correct but is no longer applicable.

Future systems should represent source dates, effective periods, amendments, and relevant relationships between authorities. Where available and properly established, this information can support temporal filtering and authority-aware retrieval. An important distinction is between the date a document was published and the period during which its legal rule applied. These dates may differ. A reliable system should not assume that the newest document is necessarily the most authoritative or applicable.

Research should evaluate whether temporal and authority-aware retrieval reduces outdated-law errors, and how systems should respond when the corpus does not contain enough information to establish current legal status.

10.8 Research direction 7: Reproducible evaluation and realistic legal benchmarks

Legal AI evaluation remains fragmented across tasks, jurisdictions, corpora, and definitions of correctness. A model may perform well on a curated benchmark but struggle with ambiguous queries, incomplete evidence, or questions that require multiple sources.

Future benchmarks should combine retrieval and generation evaluation while preserving separate scores for each stage. They should include difficult cases involving exceptions, cross-references, competing authorities, temporal changes, multilingual queries, and insufficient evidence. A strong evaluation protocol should also report dataset construction, source coverage, annotation instructions, reviewer expertise, inter-annotator agreement, baseline configurations, and uncertainty estimates. Ablation studies should isolate the contributions of hybrid retrieval, rank fusion, reranking, hierarchical indexing, knowledge graphs, and verification.

Public datasets and reproducible evaluation code can help researchers distinguish genuine architectural improvements from differences in test construction or experimental settings. Where legal or privacy restrictions prevent full data release, researchers should document the limitations and provide reproducible alternatives where possible.

10.9 Consolidated research agenda

Table 19 draws the directions together with measurable research objectives.

These research directions are interdependent. Knowledge graphs can support authority-aware retrieval, but their usefulness depends on accurate source extraction. Multilingual retrieval can improve access, but its outputs still require jurisdictional filtering. Structure-aware indexing can preserve context, while claim-level verification can assess whether the generated answer uses that context correctly. Agentic workflows can coordinate these capabilities, but only reproducible evaluation can establish whether the resulting system is more dependable. The most promising research agenda is therefore an integrated one: knowledge-graph-grounded, multilingual, structure-aware legal retrieval with explicit evidence verification and reproducible evaluation.

10.10 Concluding synthesis

The evolution of legal AI demonstrates a shift from systems focused primarily on rules, cases, and legal language understanding toward architectures that retrieve external evidence and coordinate more complex legal information workflows. Yet access to evidence is not equivalent to legal correctness. Dependable systems must preserve the structure and provenance of legal sources, recognize jurisdictional and temporal constraints, support multilingual queries, and verify that generated claims are adequately supported. Three priorities emerge from this survey:

1. represent legal knowledge more faithfully through hierarchical indexing, explicit relationships, and provenance-aware knowledge graphs;
2. make evidence retrieval more accessible and reliable through hybrid and multilingual retrieval, contextual ranking, and jurisdiction-aware filtering;
3. make generated answers more verifiable through claim-level evidence checks, authority validation, appropriate abstention, and reproducible expert evaluation.

For Pakistani legal AI, these priorities provide a concrete path toward systems that connect users' questions to applicable legal evidence without treating semantic similarity or fluent generation as proof of correctness. More broadly, they suggest that progress in legal AI should be measured not only by what a model can generate, but by whether its answers can be traced, evaluated, and justified through the legal sources on which they depend.

11 Conclusion

Legal artificial intelligence has evolved from early rule-based systems and case-based reasoning toward domain-adapted language models, retrieval-augmented generation, and increasingly sophisticated legal information workflows. Each stage has addressed limitations of earlier approaches, but the field's development also reveals a persistent distinction between producing plausible legal language and providing a legally supportable answer.

Early systems demonstrated that legal reasoning could be represented through explicit rules, structured knowledge, and comparisons between cases. These approaches offered interpretability but faced challenges in maintaining extensive legal knowledge and adapting to changing legal materials. Transformer-based language models subsequently improved the processing of legal language, while benchmarks such as LEGAL-BERT-related evaluations, LexGLUE, and LegalBench enabled more systematic assessment of specialized language understanding and legal reasoning. Nevertheless, benchmark performance alone cannot establish that a system retrieves current authority, interprets it correctly, or cites it accurately.

Retrieval-augmented generation introduced an important architectural shift by connecting language generation to external evidence. Legal RAG systems can retrieve statutory provisions, judgments, and other source materials at inference time, making answers more traceable than approaches that depend exclusively on model parameters. Research on precise legal passage retrieval, structured document drafting, and jurisdiction-specific question answering illustrates the breadth of this development. Yet RAG does not guarantee reliability: relevant passages may be omitted, retrieved fragments may lose essential context, and generated claims may exceed what the evidence supports.

The comparative analysis identifies three findings. First, retrieval quality, legal correctness, and citation support are distinct dimensions and must be evaluated separately. Second, legal document structure,

Table 19: Open challenges and measurable research objectives.

Research challenge	Proposed direction	Example evaluation
Missing cross-document relationships	Combine knowledge graphs with text retrieval.	Recall on multi-hop questions and graph-relation accuracy.
Low-resource multilingual queries	Multilingual retrieval with terminology alignment.	Language-specific Recall@k and expert-assessed correctness.
Loss of legal hierarchy	Structure-aware indexing and parent-child retrieval.	Structural extraction accuracy and downstream answer quality.
Unsupported legal claims	Claim-level evidence verification.	Claim-support precision, recall, and unsupported-claim rate.
Unreliable agentic workflows	Modular orchestration with explicit quality gates.	Correctness, failure recovery, latency, and ablation results.
Outdated or inapplicable authorities	Temporal and authority-aware retrieval.	Outdated-law error rate and applicability accuracy.
Incomparable research results	Shared benchmarks and reproducible protocols.	Cross-system results under common data and evaluation rules.
Limited auditability	Preserve provenance and traceable evidence links.	Citation resolution, provenance completeness, and audit success rate.

jurisdiction, and temporal validity are essential to determining whether retrieved evidence is applicable. Third, more complex architectures, including knowledge graphs and agentic workflows, offer opportunities to address these problems, but their value must be demonstrated through controlled evaluation rather than inferred from architectural sophistication.

These findings are particularly relevant to multilingual and jurisdiction-specific legal information retrieval. In contexts such as Pakistan, effective systems must connect user queries with the appropriate legal sources while preserving statutory hierarchy, judicial context, and source provenance. Supporting English, Urdu, and Roman Urdu adds further challenges involving terminology alignment, transliteration, and cross-lingual semantic matching. A system that handles these challenges successfully could improve access to legal information, but only if its outputs remain traceable to applicable authorities and its uncertainty is handled responsibly.

Future research should therefore prioritize structure-preserving retrieval, provenance-aware legal knowledge graphs, multilingual query processing, claim-level citation verification, and agentic workflows with explicit quality gates. Equally important are reproducible benchmarks that test the complete evidence-to-answer pipeline, including difficult cases involving exceptions, conflicting authorities, temporal changes, and insufficient evidence. Such evaluation should report retrieval effectiveness, legal correctness, citation support, latency, and computational cost without collapsing them into a single score.

Ultimately, the goal of legal AI should not be to replace legal judgment with fluent automated responses. It should be to make legal information more accessible, systematic, and auditable while preserving the distinction between retrieved evidence, model-generated interpretation, and conclusions requiring professional review. The next stage of legal AI will depend less on generation alone and more on the ability to retrieve, contextualize, verify, and reproduce evidence-supported legal reasoning.

References

Manzar Ali, Sameer Imtiaz, Farheen Qazi, Saad Khan, and Syed Muneer. AI-powered legal assistance in Pakistani law: A RAG-based framework for context-aware and reliable legal answer. *International Journal of Multidisciplinary Conference Proceedings*, 2025. doi: 10.61503/Ijmcp.v2i1.204.

Kevin D. Ashley. Reasoning with cases and hypotheticals in HYPO. *International Journal of Man-Machine Studies*, 1991. doi: 10.1016/0020-7373(91)90011-U.

Matteo Buffa, Alfio Ferrara, Sergio Picascia, Davide Riva, and Silvana Castano. Enhancing legal document building with retrieval-augmented generation. *Computer Law & Security Review*, 2025. doi: 10.1016/j.clsr.2025.106229.

Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. LEGAL-BERT: The Muppets straight out of Law School. *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020. doi: 10.18653/v1/2020.findings-emnlp.261.

Ilias Chalkidis, Abhik Jana, Dirk Hartung, Michael Bommarito, Ion Androutsopoulos, Daniel Katz, and Nikolaos Aletras. LexGLUE: A benchmark dataset for legal language understanding in English. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022. doi: 10.18653/v1/2022.acl-long.297.

Xiao Chi, Wei Wang, Ziyao Zhang, Ang Li, Yuting Huang, Yiquan Wu, Kun Kuang, Changlong Sun, Xiaozhong Liu, Fei Wu, and Minghui Xiong. Legalai research in LLM era: Data, modeling and evaluation. *Artificial Intelligence Review*, 2026. doi: 10.1007/s10462-026-11514-9.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In

Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval, 2009. doi: 10.1145/1571941.1572114.

Matthew Dahl, Varun Magesh, Mirac Suzgun, and Daniel E Ho. Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis*, 2024. doi: 10.1093/jla/laae003.

Yi Feng, Chuanyi Li, and Vincent Ng. Legal case retrieval: A survey of the state of the art. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2024. doi: 10.18653/v1/2024.acl-long.350.

Neel Guha, Julian Nyarko, Daniel E. Ho, Christopher Ré, Adam Chilton, Aditya Narayana, Alex Chohlas-Wood, Austin Peters, Brandon Waldon, Daniel N. Rockmore, Diego Zambrano, Dmitry Talisman, Enam Hoque, Faiz Surani, Frank Fagan, Galit Sarfaty, Gregory M. Dickinson, Haggai Porat, Jason Hegland, Jessica Wu, Joe Nudell, Joel Niklaus, John Nay, Jonathan H. Choi, Kevin Tobia, Margaret Hagan, Megan Ma, Michael Livermore, Nikon Rasumov-Rahe, Nils Holzenberger, Noam Kolt, Peter Henderson, Sean Rehaag, Sharad Goel, Shang Gao, Spencer Williams, Sunny Gandhi, Tom Zur, Varun Iyer, and Zehua Li. LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. *arXiv preprint*, 2023. doi: arXiv:2308.11462.

Ali Hamza, Maaz Bin Rizwan, Maemoona Kayani, and Shafa-at Ali Sheikh. Legalyze: An open-source structure-aware agentic RAG platform for high-precision legal intelligence. *Expert Systems*, 2026. doi: 10.1111/exsy.70415.

Joshua Kelsall, Xingwei Tan, Aislinn Bergin, Jiahong Chen, Maria Waheed, Tom Sorell, Rob Procter, Maria Liakata, Jenny Chim, and Serene Chi. A rapid evidence review of evaluation techniques for large language models in legal use cases: Trends, gaps, and recommendations for future research. *AI & Society*, 2025. doi: 10.1007/s00146-025-02741-9.

Varun Magesh, Faiz Surani, Matthew Dahl, Mirac Suzgun, Christopher D. Manning, and Daniel E. Ho. Hallucination-free? assessing the reliability of leading AI legal research tools. *Journal of Empirical Legal Studies*, 2025. doi: 10.1111/jels.12413.

Nicholas Pipitone and Ghita Houir Alami. LegalBench-RAG: A benchmark for retrieval-augmented generation in the legal domain. *arXiv preprint*, 2024. doi: arXiv:2408.10343.

Damith Premasiri, Tharindu Ranasinghe, Ruslan Mitkov, Mo El-Haj, and Ingo Frommholz. Survey on legal information extraction: current status and open challenges. *Knowledge and Information Systems*, 2025. doi: 10.1007/s10115-025-02600-5.

Wenqi Ren, Lixing Shen, Jiang Zhu, Shaohui Li, Yinxia Hong, Jiawei Wang, and Da Cao. Automated data synthesis and retrieval-augmented generation for legal large language models. *Knowledge-Based Systems*, 2026. doi: 10.1016/j.knosys.2026.116267.